

数学物理方程



9 习题

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习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = \sin \frac{3\pi}{2l} x, u_t(x, 0) = \sin \frac{5\pi}{2l} x, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

2齐

解 第1步 确定特征值/特征函数

边界已经齐次, [1, 2]型边界

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin \left(\frac{n\pi}{l} x \right), (n = 1, 2, 3 \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos \left(\frac{n\pi}{l} x \right), (n = 0, 1, 2, 3 \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n = 0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n = 1, 2, 3 \dots)$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = \sin \frac{3\pi}{2l} x, u_t(x, 0) = \sin \frac{5\pi}{2l} x, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

2齐

第2步 正交分解

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$$

Fourier级数展开

设原问题有分离变量的形式解

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \sin \left[\frac{(2n+1)\pi}{2l} x \right]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, (n = 0, 1, 2, 3 \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, (n = 1, 2, 3 \dots)$$

自由项/初始条件按特征函数系展开

$$u(x, 0) = \sin \frac{3\pi}{2l} x$$

 $n=1$

初始分布与特征函数同型

$$= \sum_{n=0}^{\infty} \phi_n X_n(x) = \sum_{n=0}^{\infty} \phi_n \sin \left[\frac{(2n+1)\pi}{2l} x \right]$$



$$\phi_n = \begin{cases} \frac{2}{l} \int_0^l \sin \frac{3\pi}{2l} x \sin \frac{3\pi}{2l} x \cdot dx = 1, & (n=1) \\ 0, & (n \neq 1) \end{cases}$$

$$u_x(x, 0) = \sin \frac{5\pi}{2l} x$$

 $n=2$

$$= \sum_{n=0}^{\infty} \psi_n X_n(x) = \sum_{n=0}^{\infty} \psi_n \sin \left[\frac{(2n+1)\pi}{2l} x \right]$$



$$\psi_n = \begin{cases} \frac{2}{l} \int_0^l \sin \frac{5\pi}{2l} x \sin \frac{5\pi}{2l} x \cdot dx = 1, & (n=2) \\ 0, & (n \neq 2) \end{cases}$$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = \sin \frac{3\pi}{2l} x, u_t(x, 0) = \sin \frac{5\pi}{2l} x, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

2齐

第3步 建立初值问题ODE

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3, \dots)$$

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \sin \left[\frac{(2n+1)\pi}{2l} x \right]$$

假设形式解可逐项求导, 代入原PDE

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) X_n''(x) = 0$$

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) [-\lambda_n X_n(x)] = 0$$

$$\sum_{n=0}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = 0$$

$$T_n''(t) + a^2 \lambda_n T_n(t) = 0$$

用原PDE初始条件→初值问题ODE初始条件

$$u(x, 0) = \sum_{n=0}^{\infty} T_n(0) X_n(x) = \sum_{n=0}^{\infty} \varphi_n X_n(x)$$

$$u_t(x, 0) = \sum_{n=0}^{\infty} T_n'(0) X_n(x) = \sum_{n=0}^{\infty} \psi_n X_n(x)$$

建立了初值问题ODE

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases} \quad (n = 1, 2)$$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = \sin \frac{3\pi}{2l} x, u_t(x, 0) = \sin \frac{5\pi}{2l} x, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

2齐

第4步 求初值问题

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3, \dots)$$

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases} \quad (n=1, 2) \quad \text{二阶线性齐次ODE}$$

$$\varphi_n = \begin{cases} 1, & (n=1) \\ 0, & (n \neq 1) \end{cases}$$

$$\psi_n = \begin{cases} 1, & (n=2) \\ 0, & (n \neq 2) \end{cases}$$

各阶导数

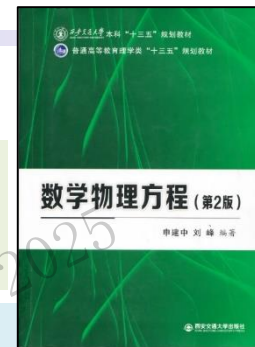
$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t)$$

$$T_n'(t) = -C_{1n} a\sqrt{\lambda_n} \sin(a\sqrt{\lambda_n}t) + C_{2n} a\sqrt{\lambda_n} \cos(a\sqrt{\lambda_n}t)$$

根据初始条件确定系数

$$\begin{cases} T_n(0) = C_{1n} = \varphi_n \\ T_n'(0) = C_{2n} a\sqrt{\lambda_n} = \psi_n \end{cases} \quad \rightarrow \quad \begin{cases} C_{1n} = \varphi_n \\ C_{2n} = \frac{\psi_n}{a\sqrt{\lambda_n}} \end{cases} \quad (n=1, 2)$$

$$T_n(t) = \varphi_n \cos\left(\frac{(2n+1)\pi a}{2l} t\right) + \psi_n \frac{2l}{(2n+1)\pi a} \sin\left(\frac{(2n+1)\pi a}{2l} t\right), \quad (n=1, 2)$$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = \sin \frac{3\pi}{2l} x, u_t(x, 0) = \sin \frac{5\pi}{2l} x, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

2齐

$$\varphi_n = \begin{cases} 1, & (n=1) \\ 0, & (n \neq 1) \end{cases}$$

$$\psi_n = \begin{cases} 1, & (n=2) \\ 0, & (n \neq 2) \end{cases}$$

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n=0, 1, 2, 3, \dots)$$

$$T_n(t) = \varphi_n \cos \left(\frac{(2n+1)\pi a}{2l} t \right) + \psi_n \frac{2l}{(2n+1)\pi a} \sin \left(\frac{(2n+1)\pi a}{2l} t \right)$$

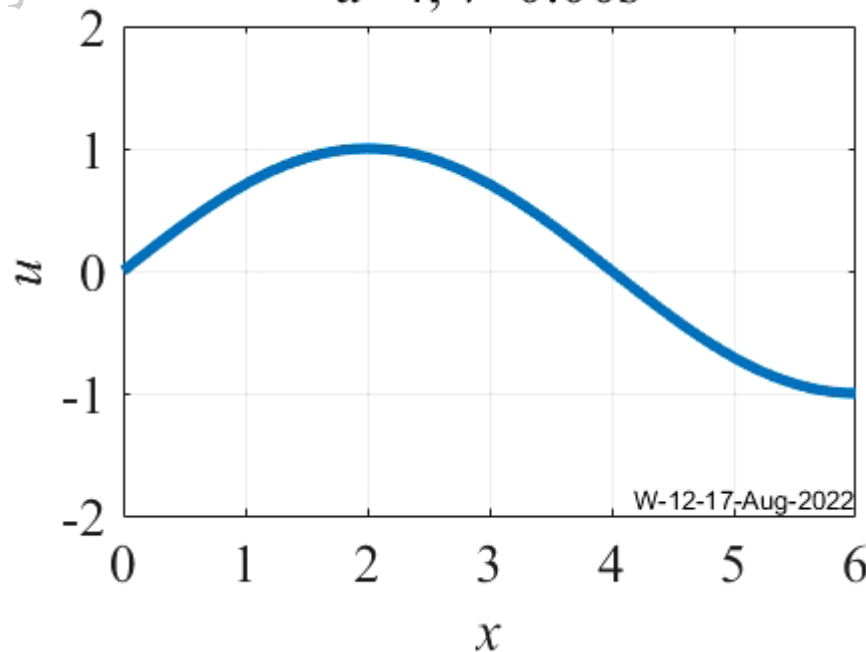
$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x)$$

$$= T_1(t) X_1(x) + T_2(t) X_2(x)$$

$$= T_1(t) \sin \frac{3\pi}{2l} x + T_2(t) \sin \frac{5\pi}{2l} x$$

$$= \cos \frac{3\pi a}{2l} t \cdot \sin \frac{3\pi}{2l} x + \frac{2l}{5\pi a} \sin \frac{5\pi a}{2l} t \cdot \sin \frac{5\pi}{2l} x$$

若初始分布恰好为某个特征函数, 可仅针对该n值对应的特征函数展开

 $a=4, \tau=0.00s$


W-12-17-Aug-2022



习题

例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

导热/一维/齐次

2非齐

1非齐

解 第1步 边界条件齐次化

设 $u(x, t) = v(x, t) + w(x, t)$ 可取 $w(x, t) = 1 \cdot (x - \pi) + \pi = x$ 各阶 $w_t = 0$ 导数 $w_x = 1, w_{xx} = 0$

改造原PDE

$$\begin{aligned} u_t - a^2 u_{xx} &= \cos \frac{x}{2} \\ v_t + w_t - a^2 v_{xx} - a^2 w_{xx} &= \cos \frac{x}{2} \\ v_t - a^2 v_{xx} &= \cos \frac{x}{2} \end{aligned}$$

改造初始条件

$$\begin{aligned} u(x, 0) &= 0 \\ v(x, 0) + w(x, 0) &= 0 \\ v(x, 0) + x &= 0 \\ v(x, 0) &= -x \end{aligned}$$



原定解问题可转化为

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

边界条件齐次化常用辅助函数 $u(x, t) = v(x, t) + w(x, t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0, t) = g_1(t), u(l, t) = g_2(t)$	$w(x, t) = \frac{g_2(t) - g_1(t)}{l} x + g_1(t)$
[2,1]	$u_x(0, t) = g_1(t), u(l, t) = g_2(t)$	$w(x, t) = g_1(t)(x - l) + g_2(t)$
[1,2]	$u(0, t) = g_1(t), u_x(l, t) = g_2(t)$	$w(x, t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0, t) = g_1(t), u_x(l, t) = g_2(t)$	$w(x, t) = \frac{g_2(t) - g_1(t)}{2l} x^2 + g_1(t)x$



例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

第2步 确定特征值/特征函数

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$$

本例中 $l = \pi$

$$\lambda_n = \left(\frac{2n+1}{2} \right)^2, X_n(x) = \cos \left(\frac{2n+1}{2} x \right), (n = 0, 1, 2, 3 \dots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin \left(\frac{n\pi}{l} x \right), (n = 1, 2, 3 \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos \left(\frac{n\pi}{l} x \right), (n = 0, 1, 2, 3 \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n = 0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n = 1, 2, 3 \dots)$

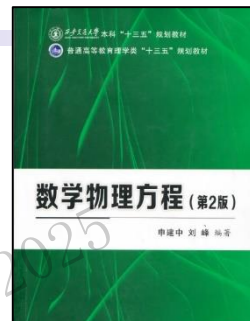
习题

导热/一维/齐次

2非齐

1非齐

习题2, 6(1)



例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

第3步 正交分解

$$\lambda_n = \left(\frac{2n+1}{2} \right)^2, X_n(x) = \cos \left(\frac{2n+1}{2} x \right), (n = 0, 1, 2, 3 \dots)$$

设原问题有分离变量的形式解

Fourier级数展开

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos \left(\frac{2n+1}{2} x \right)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, (n = 0, 1, 2, 3 \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, (n = 1, 2, 3 \dots)$$

自由项/初始条件按特征函数系展开

$$\cos \frac{x}{2} = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \cos \left(\frac{2n+1}{2} x \right)$$

$$f_n = \frac{2}{l} \int_0^l \cos \frac{x}{2} \cos \frac{2n+1}{2} x \cdot dx = \begin{cases} 1, & (n = 0) \\ 0, & (n \neq 0) \end{cases}$$

$$v(x, 0) = -x = \sum_{n=0}^{\infty} \varphi_n \cos \left(\frac{2n+1}{2} x \right)$$

$$\varphi_n = \frac{2}{l} \int_0^l (-x) \cos \left(\frac{2n+1}{2} x \right) \cdot dx = -\frac{2}{\pi} \left[\frac{2}{2n+1} x \sin \frac{2n+1}{2} x + \frac{4}{(2n+1)^2} \cos \frac{2n+1}{2} x \right]_0^{\pi} = \frac{4(-1)^{n+1}}{2n+1} + \frac{8}{(2n+1)^2 \pi}$$



例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

第4步 建立初值问题ODE

$$v(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos\left(\frac{2n+1}{2}x\right)$$

假设形式解可逐项求导, 代入原PDE

$$\begin{aligned} \sum_{n=0}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) X_n''(x) &= \sum_{n=0}^{\infty} f_n X_n(x) \\ \sum_{n=0}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) [-\lambda_n X_n(x)] &= \sum_{n=0}^{\infty} f_n X_n(x) \\ \sum_{n=0}^{\infty} [T_n'(t) + a^2 \lambda_n T_n(t)] X_n(x) &= \sum_{n=0}^{\infty} f_n X_n(x) \\ T_n'(t) + a^2 \lambda_n T_n(t) &= f_n \end{aligned}$$

$$\lambda_n = \left(\frac{2n+1}{2}\right)^2, X_n(x) = \cos\left(\frac{2n+1}{2}x\right), (n=0, 1, 2, 3, \dots)$$

用原PDE初始条件→初值问题ODE初始条件

$$v(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = -x = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

建立了初值问题ODE

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n \end{cases}$$



例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

第5步 求初值问题

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n \end{cases}$$

二阶线性非齐次ODE, 通解的形式为

$$T_n(t) = C_{1n} \exp(-a^2 \lambda_n t) + C_{2n}$$

$$T_n'(t) = -a^2 \lambda_n C_{1n} \exp(-a^2 \lambda_n t)$$

$$-a^2 \lambda_n C_{1n} \exp(-a^2 \lambda_n t) + a^2 \lambda_n C_{1n} \exp(-a^2 \lambda_n t) + a^2 \lambda_n C_{2n} = f_n$$

比较系数得 $C_{2n} = \frac{f_n}{a^2 \lambda_n}$

由初始条件 $T_n(0) = \varphi_n = C_{1n} + \frac{f_n}{a^2 \lambda_n} \Rightarrow C_{1n} = \varphi_n - \frac{f_n}{a^2 \lambda_n}$

$$\lambda_n = \left(\frac{2n+1}{2} \right)^2, X_n(x) = \cos\left(\frac{2n+1}{2} x \right), (n = 0, 1, 2, 3 \dots)$$

$$f_n = \frac{2}{l} \int_0^l \cos \frac{x}{2} \cos \frac{2n+1}{2} x \cdot dx = \begin{cases} 1, & (n=0) \\ 0, & (n \neq 0) \end{cases}$$

$$\varphi_n = \frac{2}{l} \int_0^l (-x) \cos\left(\frac{2n+1}{2} x \right) \cdot dx = \frac{4(-1)^{n+1}}{2n+1} + \frac{8}{(2n+1)^2 \pi}$$

初值ODE的解

$$T_n(t) = \left(\varphi_n - \frac{f_n}{a^2 \lambda_n} \right) \exp(-a^2 \lambda_n t) + \frac{f_n}{a^2 \lambda_n}$$



例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

第6步 回代

$$f_n = \frac{2}{l} \int_0^l \cos \frac{x}{2} \cos \frac{2n+1}{2} x \cdot dx = \begin{cases} 1, & (n=0) \\ 0, & (n \neq 0) \end{cases}$$

$$\lambda_n = \left(\frac{2n+1}{2} \right)^2, X_n(x) = \cos \left(\frac{2n+1}{2} x \right), (n=0, 1, 2, 3 \dots)$$

$$\varphi_n = \frac{2}{l} \int_0^l (-x) \cos \left(\frac{2n+1}{2} x \right) \cdot dx = \frac{4(-1)^{n+1}}{2n+1} + \frac{8}{(2n+1)^2 \pi}$$

↓ n 是否=0

$$T_n(t) = \left(\varphi_n - \frac{f_n}{a^2 \lambda_n} \right) \exp(-a^2 \lambda_n t) + \frac{f_n}{a^2 \lambda_n}$$

边界条件齐次
化辅助函数

$n=0$ 单列

$$u(x, t) = w(x, t) + v(x, t)$$

$$= x + T_0(t) \cos \frac{x}{2} + \sum_{n=1}^{\infty} T_n(t) \cos \left(\frac{2n+1}{2} x \right)$$

$$= x + \left[\left(\varphi_0 - \frac{4}{a^2} \right) \exp \left(-\frac{a^2}{4} t \right) + \frac{4}{a^2} \right] \cos \frac{x}{2} + \sum_{n=1}^{\infty} \varphi_n \exp(-a^2 \lambda_n t) \cos \left(\frac{2n+1}{2} x \right)$$



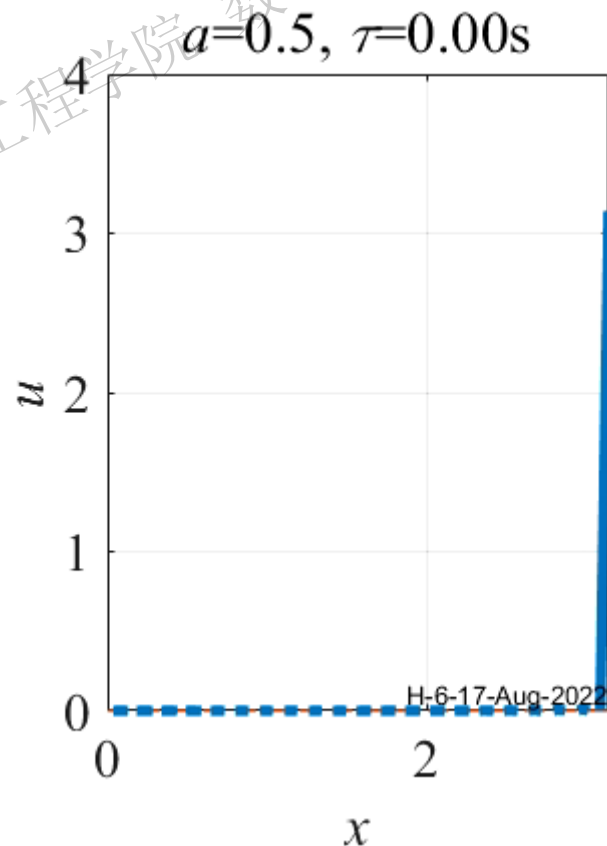
例

$$\begin{cases} u_t - a^2 u_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 1, u(\pi, t) = \pi, & t \geq 0 \\ u(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = \cos \frac{x}{2}, & 0 < x < \pi, t > 0 \\ v_x(0, t) = 0, v(\pi, t) = 0, & t \geq 0 \\ v(x, 0) = -x, & 0 \leq x \leq \pi \end{cases}$$

$$\lambda_n = \left(\frac{2n+1}{2} \right)^2, X_n(x) = \cos \left(\frac{2n+1}{2} x \right), (n = 0, 1, 2, 3 \dots) \quad \varphi_n = \frac{2}{l} \int_0^l (-x) \cos \left(\frac{2n+1}{2} x \right) dx = \frac{4(-1)^{n+1}}{2n+1} + \frac{8}{(2n+1)^2 \pi}$$

$$u(x, t) = x + \left[\left(\varphi_0 - \frac{4}{a^2} \right) \exp \left(-\frac{a^2}{4} t \right) + \frac{4}{a^2} \right] \cos \frac{x}{2} + \sum_{n=1}^{\infty} \varphi_n \exp(-a^2 \lambda_n t) \cos \left(\frac{2n+1}{2} x \right)$$





习题

例

$$\begin{cases} u_t - a^2 u_{xx} + b^2 u = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

解 第1步 确定特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3\cdots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l} x \right], (n=0, 1, 2, 3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l} x \right], (n=0, 1, 2, 3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x \right), (n=0, 1, 2, 3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\begin{aligned} \lambda_0 &= 0, \Phi_0(\theta) = 1, (n=0) \\ \lambda_n &= n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3\cdots) \end{aligned}$



习题

例

$$\begin{cases} u_t - a^2 u_{xx} + b^2 u = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次	
1齐	1齐

第2步 正交分解

设原问题有分离变量的形式解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x\right)$$

自由项/初始条件按特征函数系展开

$$u(x, 0) = \varphi(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l} x\right)$$

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x\right), (n=1, 2, 3, \dots)$$

Fourier级数展开

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n=0, 1, 2, 3, \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$



习题

例

$$\begin{cases} u_t - a^2 u_{xx} + b^2 u = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次	
1齐	1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3, \dots)$$

第3步 建立初值问题ODE

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x \right)$$

用原PDE初始条件→初值问题ODE初始条件

假设形式解可逐项求导, 代入原PDE

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \varphi(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

$$\sum_{n=1}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) X_n''(x) + b^2 \sum_{n=1}^{\infty} T_n(t) X_n(x) = 0$$

$$\sum_{n=1}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) [-\lambda_n X_n(x)] + b^2 \sum_{n=1}^{\infty} T_n(t) X_n(x) = 0$$

$$\sum_{n=1}^{\infty} [T_n'(t) + (a^2 \lambda_n + b^2) T_n(t)] X_n(x) = 0$$

$$T_n'(t) + (a^2 \lambda_n + b^2) T_n(t) = 0$$

建立了初值问题ODE

$$\begin{cases} T_n'(t) + (a^2 \lambda_n + b^2) T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$



习题

例

$$\begin{cases} u_t - a^2 u_{xx} + b^2 u = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次	
1齐	1齐

第4步 求初值问题

$$\begin{cases} T_n'(t) + (a^2 \lambda_n + b^2) T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

二阶线性齐次ODE, 通解的形式为

$$T_n(t) = C_{1n} \exp[-(a^2 \lambda_n + b^2)t]$$

由初始条件

$$T_n(0) = \varphi_n = C_{1n} \Rightarrow C_{1n} = \varphi_n$$

初值ODE的解

$$T_n(t) = \varphi_n \exp[-(a^2 \lambda_n + b^2)t]$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

原定解问题的解可表示为

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right) \\ &= \sum_{n=1}^{\infty} \varphi_n \exp[-(a^2 \lambda_n + b^2)t] \sin\left(\frac{n\pi}{l}x\right) \end{aligned}$$

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx$$



习题

例

$$\begin{cases} u_t - a^2 u_{xx} + b^2 u = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

讨论

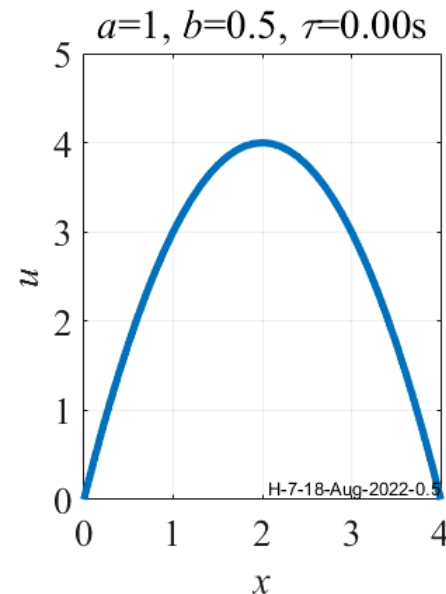
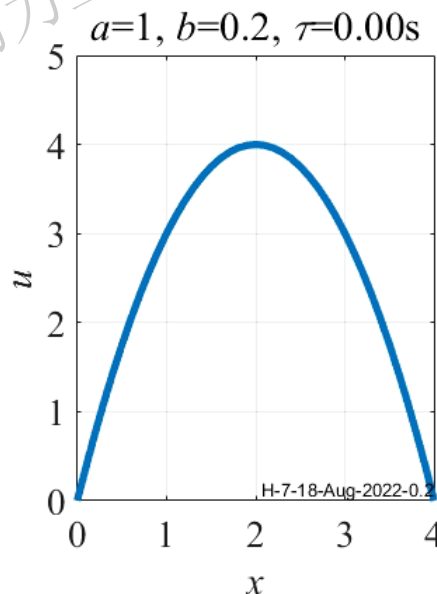
$$u(x, t) = \sum_{n=1}^{\infty} \varphi_n \exp[-(a^2 \lambda_n + b^2)t] \sin\left(\frac{n\pi}{l}x\right)$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

设初始分布

$$\varphi(x) = x(l-x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right)$$

$$\begin{aligned} \varphi_n &= \frac{2}{l} \int_0^l x(l-x) \sin\left(\frac{n\pi}{l}x\right) dx \\ &= \frac{2}{l} \left[l \int_0^l x \sin \frac{n\pi x}{l} \cdot dx - \int_0^l x^2 \sin \frac{n\pi x}{l} \cdot dx \right] \\ &= \frac{4l^2}{\pi^3} \cdot \frac{1 - \cos n\pi}{n^3} = \frac{4l^2}{\pi^3} \cdot \frac{1 - (-1)^n}{n^3} \end{aligned}$$

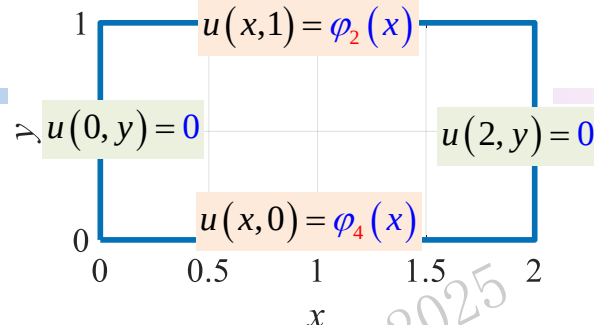


习题

例

$$\begin{cases} u_{xx} + u_{yy} = A, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(l, y) = 0, & 0 \leq y \leq h \\ u(x, 0) = \varphi_4(x), \quad u(x, h) = \varphi_2(x), & 0 \leq x \leq l \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1非齐	1非齐



解 第1步 特征值问题

方程非齐次/x边界[1,1]型 → 特征值/特征函数为

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), \quad (n=1, 2, 3, \dots)$$

第2步 正交分解

方程的解/自由项/ y边界按照特征函数系展开

$$u(x, y) = \sum_{n=1}^{\infty} Y_n(y) X_n(x) = \sum_{n=1}^{\infty} Y_n(y) \sin\left(\frac{n\pi}{l}x\right)$$

$$A = \sum_{n=1}^{\infty} f_n X_n(x) \quad f_n = \frac{2}{l} \int_0^l A \sin\left(\frac{n\pi}{l}x\right) dx = \frac{2A}{n\pi} [1 - \cos n\pi]$$

$$\varphi_4(x) = \sum_{n=1}^{\infty} \varphi_{4n}(x) X_n(x) \quad \varphi_{2n} = \frac{2}{l} \int_0^l \varphi_2(x) \sin\left(\frac{n\pi}{l}x\right) dx, \quad (n=1, 2, 3, \dots)$$

$$\varphi_2(x) = \sum_{n=1}^{\infty} \varphi_{2n}(x) X_n(x) \quad \varphi_{4n} = \frac{2}{l} \int_0^l \varphi_4(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

Fourier级数展开

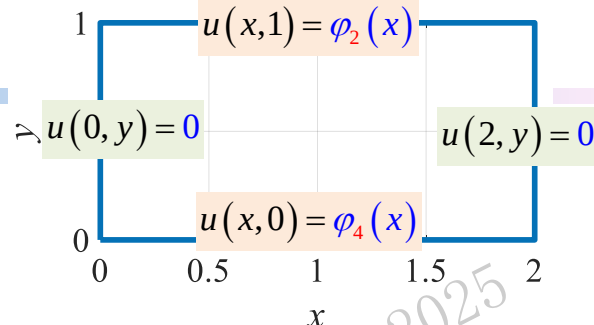
$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \\ a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n=0, 1, 2, 3, \dots) \\ b_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots) \end{aligned}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = A, & 0 < x < l, & 0 < y < h \\ u(0, y) = 0, & u(l, y) = 0, & 0 \leq y \leq h \\ u(x, 0) = \varphi_4(x), & u(x, h) = \varphi_2(x), & 0 \leq x \leq l \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1非齐	1非齐



第3步 建立y方向ODE

正交分解形式代入原方程

$$\sum_{n=1}^{\infty} Y_n(y) X_n''(x) + \sum_{n=1}^{\infty} Y_n''(y) X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\sum_{n=1}^{\infty} Y_n(y) [-\lambda_n X_n(x)] + \sum_{n=1}^{\infty} Y_n''(y) X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\sum_{n=1}^{\infty} [Y_n''(y) - \lambda_n Y_n(y)] X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$Y_n''(y) - \lambda_n Y_n(y) = f_n$$

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l} x \right), \quad (n=1, 2, 3, \dots)$$

根据下底面边界 $u(x, 0) = \varphi_4(x)$

$$\sum_{n=1}^{\infty} Y_n(0) X_n(x) = \sum_{n=1}^{\infty} \varphi_{4n} X_n(x)$$

$$Y_n(0) = \varphi_{4n}$$

根据上底面边界 $u(x, h) = \varphi_2(x)$

$$\sum_{n=1}^{\infty} Y_n(h) X_n(x) = \sum_{n=1}^{\infty} \varphi_{2n} X_n(x)$$

$$Y_n(h) = \varphi_{2n}$$

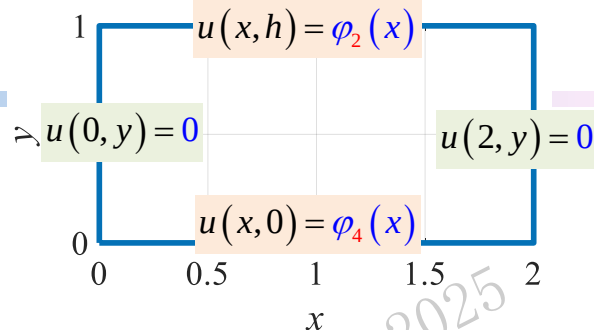
$$\begin{cases} Y_n''(y) - \lambda_n Y_n(y) = f_n & 0 < y < h \\ Y_n(0) = \varphi_{4n}, Y_n(h) = \varphi_{2n}, & 0 \leq y \leq h \end{cases}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = A, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(l, y) = 0, & 0 \leq y \leq h \\ u(x, 0) = \varphi_4(x), \quad u(x, h) = \varphi_2(x), & 0 \leq x \leq l \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1非齐	1非齐



第4步 解y方向ODE

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), \quad (n=1, 2, 3, \dots)$$

$$\begin{cases} Y_n''(y) - \lambda_n Y_n(y) = f_n & 0 < y < h \\ Y_n(0) = \varphi_{4n}, \quad Y_n(h) = \varphi_{2n}, & 0 \leq y \leq h \end{cases}$$

通解形式为

$$Y(y) = C_{1n} \sinh\left(\frac{n\pi}{l}y\right) + C_{2n} \cosh\left(\frac{n\pi}{l}y\right) + \bar{Y}$$

假设特解为常数 $\bar{Y} = C$ 代入ODE

$$0 - \lambda_n C = f_n \quad \bar{Y} = C = -\frac{f_n}{\lambda_n}$$

通解可表示为

$$Y(y) = C_{1n} \sinh\left(\frac{n\pi}{l}y\right) + C_{2n} \cosh\left(\frac{n\pi}{l}y\right) - \frac{f_n}{\lambda_n}$$

根据边界条件

$$Y(0) = \varphi_{4n} = C_{2n} - \frac{f_n}{\lambda_n}$$

$$Y(h) = \varphi_{2n} = C_{1n} \sinh\left(\frac{n\pi}{l}h\right) + C_{2n} \cosh\left(\frac{n\pi}{l}h\right) - \frac{f_n}{\lambda_n}$$

$$\text{解得} \quad C_{2n} = \varphi_{4n} + \frac{f_n}{\lambda_n}$$

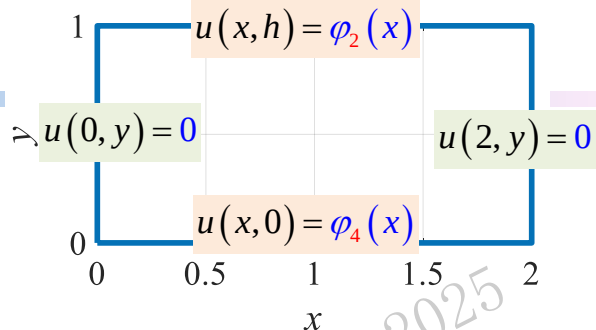
$$C_{1n} = \frac{\varphi_{2n} - \varphi_{4n} \cosh\left(\frac{n\pi}{l}h\right) + \frac{f_n}{\lambda_n} \left[1 - \cosh\left(\frac{n\pi}{l}h\right)\right]}{\sinh\left(\frac{n\pi}{l}h\right)}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = A, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(l, y) = 0, & 0 \leq y \leq h \\ u(x, 0) = \varphi_4(x), \quad u(x, h) = \varphi_2(x), & 0 \leq x \leq l \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1非齐	1非齐



第5步 回代

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), \quad (n=1, 2, 3, \dots)$$

$$u(x, y) = \sum_{n=1}^{\infty} \left[C_{1n} \sinh\left(\frac{n\pi}{l}y\right) + C_{2n} \cosh\left(\frac{n\pi}{l}y\right) - \frac{f_n}{\lambda_n} \right] \sin\left(\frac{n\pi}{l}x\right)$$

$$C_{1n} = \frac{\varphi_{2n} - \varphi_{4n} \cosh\left(\frac{n\pi}{l}h\right) + \frac{f_n}{\lambda_n} \left[1 - \cosh\left(\frac{n\pi}{l}h\right) \right]}{\sinh\left(\frac{n\pi}{l}h\right)}, \quad C_{2n} = \varphi_{4n} + \frac{f_n}{\lambda_n}$$

$$f_n = \frac{2}{l} \int_0^l A \sin\left(\frac{n\pi}{l}x\right) dx = \frac{2A}{n\pi} [1 - \cos n\pi]$$

$$\varphi_{2n} = \frac{2}{l} \int_0^l \varphi_2(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$

$$\varphi_{4n} = \frac{2}{l} \int_0^l \varphi_4(x) \sin \frac{n\pi x}{l} \cdot dx$$

可被视为矩形区域上
Laplace方程的通解

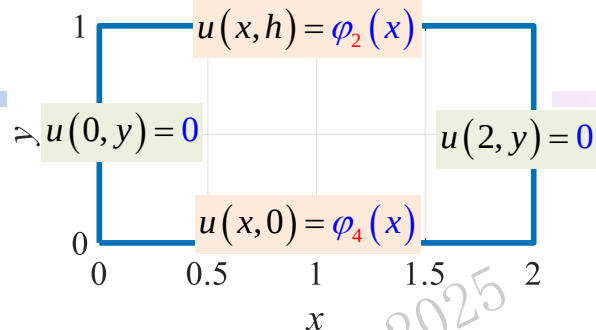
习题

例

$$\begin{cases} u_{xx} + u_{yy} = A, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(l, y) = 0, & 0 \leq y \leq h \\ u(x, 0) = \varphi_4(x), \quad u(x, h) = \varphi_2(x), & 0 \leq x \leq l \end{cases}$$

Laplace/二维/齐次

x边界	1齐	1齐
y边界	1非齐	1非齐



讨论

$$u(x, 0) = \varphi_4(x) = 0, \quad u(x, h) = \varphi_2(x) = x(x-l), \quad f = 0$$

解可简化为

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l} x \right), \quad (n=1, 2, 3, \dots)$$

$$Y(y) = C_{1n} \sinh\left(\frac{n\pi}{l} y \right)$$

$$C_{1n} = \frac{\varphi_{2n}}{\sinh\left(\frac{n\pi}{l} h \right)} \quad \rightarrow \quad \varphi_{2n} = \frac{2}{l} \int_0^l \varphi_2(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$

$$= \frac{2}{l} \int_0^l x(x-l) \sin \frac{n\pi x}{l} \cdot dx = \frac{2}{l} \int_0^l x^2 \sin \frac{n\pi x}{l} \cdot dx - 2 \int_0^l x \sin \frac{n\pi x}{l} \cdot dx = \frac{4l^2}{n^3 \pi^3} [(-1)^n - 1]$$

$$\frac{2}{l} \int_0^l x^2 \sin \frac{n\pi x}{l} \cdot dx \quad \textcircled{1}$$

$$= \frac{2}{l} \left[-\frac{l}{n\pi} x^2 \cos \frac{n\pi}{l} x + 2 \left(\frac{l}{n\pi} \right)^3 \cos \frac{n\pi}{l} x \right]_0^l$$

$$= -\frac{2l^2}{n\pi} \cos(n\pi) + \frac{4l^2}{n^3 \pi^3} \cos(n\pi) - \frac{4l^2}{n^3 \pi^3}$$

$$-2 \int_0^l x \sin \frac{n\pi x}{l} \cdot dx \quad \textcircled{2}$$

$$= -2 \left[-\frac{l}{n\pi} x \cos \frac{n\pi}{l} x + \left(\frac{l}{n\pi} \right)^2 \sin \frac{n\pi}{l} x \right]_0^l = \frac{2l^2}{n\pi} \cos n\pi$$

$$u(x, y) = \sum_{n=1}^{\infty} \left[C_{1n} \sinh\left(\frac{n\pi}{l} y \right) + C_{2n} \cosh\left(\frac{n\pi}{l} y \right) - \frac{f_n}{\lambda_n} \right] \sin\left(\frac{n\pi}{l} x \right)$$

$$\rightarrow u(x, y) = \frac{4l^2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^3} \cdot \frac{\sinh\left(\frac{n\pi}{l} y \right)}{\sinh\left(\frac{n\pi}{l} h \right)} \sin\left(\frac{n\pi}{l} x \right)$$

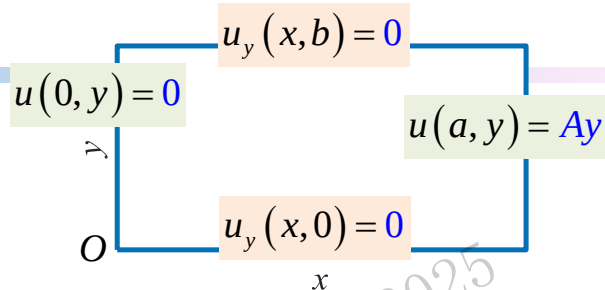
习题

例

$$\begin{cases} u_{xx} + u_{yy} = x, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(a, y) = Ay, & 0 \leq y \leq b \\ u_y(x, 0) = 0, \quad u_y(x, b) = 0, & 0 \leq x \leq a \end{cases}$$

Poisson/二维/非齐次

x边界	1齐	1非齐
y边界	2齐	2齐



解 第1步 特征值问题 方程非齐次/y边界[2,2]型→特征值/特征函数为 习题2, 7(1)

特征函数是y向的

$$\lambda_n = \left(\frac{n\pi}{b}\right)^2, \quad Y_n(x) = \cos\left(\frac{n\pi}{b}y\right), \quad (n = 0, 1, 2, 3, \dots)$$

特征函数含有1



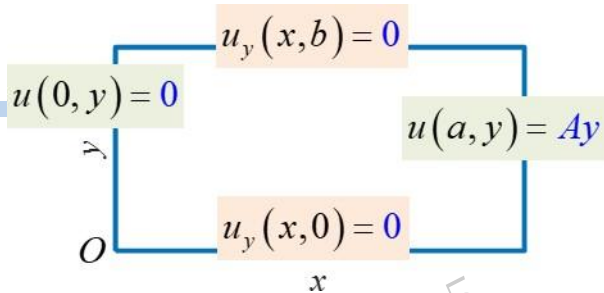
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n = 1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, \quad X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n = 0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, \quad X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n = 0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n = 0, 1, 2, 3, \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, (n = 0)$ $\lambda_n = n^2, \quad \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n = 1, 2, 3, \dots)$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = x, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(a, y) = Ay, & 0 \leq y \leq b \\ u_y(x, 0) = 0, \quad u_y(x, b) = 0, & 0 \leq x \leq a \end{cases}$$

Poisson/二维/非齐次		
x边界	1齐	1非齐
y边界	2齐	2齐



第2步 正交分解

$$\lambda_n = \left(\frac{n\pi}{b}\right)^2, \quad Y_n(x) = \cos\left(\frac{n\pi}{b}y\right), \quad (n=0, 1, 2, 3, \dots)$$

设原定解问题有分离变量的形式解

$$u(x, y) = \sum_{n=0}^{\infty} X_n(x) Y_n(y) = \sum_{n=0}^{\infty} X_n(x) \cos\left(\frac{n\pi}{b}y\right)$$

自由项按特征函数系展开

$$x = \sum_{n=0}^{\infty} f_n Y_n(y) = \sum_{n=0}^{\infty} f_n \cos\left(\frac{n\pi}{b}y\right) \begin{cases} n=0 & f_0 = \frac{1}{2} \cdot \frac{2}{b} \int_0^b x \cdot 1 \cdot dy = \frac{x}{b} \cdot b = x \\ n>0 & f_n = \frac{2}{b} \int_0^b x \cos\left(\frac{n\pi}{b}y\right) dy = \frac{2x}{b} \frac{b}{n\pi} \sin\left(\frac{n\pi}{b}y\right) \Big|_0^b = 0 \end{cases}$$

x方向右边界按特征函数系展开

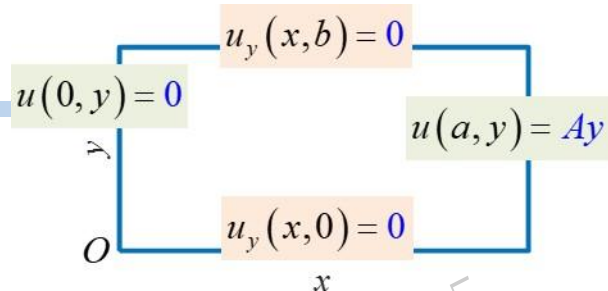
$$Ay = \sum_{n=0}^{\infty} \psi_n Y_n(y) = \sum_{n=0}^{\infty} \psi_n \cos\left(\frac{n\pi}{b}y\right) \begin{cases} n=0 & \psi_0 = \frac{1}{2} \cdot \frac{2}{b} \int_0^b Ay \cdot 1 \cdot dy = \frac{A}{b} \cdot \frac{b^2}{2} = \frac{Ab}{2} \\ n>0 & \psi_n = \frac{2}{b} \int_0^b Ay \cos\left(\frac{n\pi}{b}y\right) dy \\ & = \frac{2A}{b} \left[\frac{b}{n\pi} y \sin \frac{n\pi}{b} y + \frac{b^2}{n^2 \pi^2} \cos \frac{n\pi}{b} y \right]_0^b = \frac{2[(-1)^n - 1]}{n^2 \pi^2} Ab \end{cases}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = x, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(a, y) = Ay, & 0 \leq y \leq b \\ u_y(x, 0) = 0, \quad u_y(x, b) = 0, & 0 \leq x \leq a \end{cases}$$

Poisson/二维/非齐次		
x边界	1齐	1非齐
y边界	2齐	2齐



第3步 建立x方向ODE

$$u(x, y) = \sum_{n=0}^{\infty} X_n(x) Y_n(y) = \sum_{n=0}^{\infty} X_n(x) \cos\left(\frac{n\pi}{b} y\right)$$

设形式解可逐项求导

$$\sum_{n=0}^{\infty} X_n''(x) Y_n(y) + \sum_{n=0}^{\infty} X_n(x) Y_n''(y) = \sum_{n=0}^{\infty} f_n Y_n(y)$$

PDE的x方向边界条件→ODE边界条件

$$\sum_{n=0}^{\infty} X_n''(x) Y_n(y) + \sum_{n=0}^{\infty} X_n(x) [-\lambda_n Y_n(y)] = \sum_{n=0}^{\infty} f_n Y_n(y)$$

$$u(0, y) = \sum_{n=0}^{\infty} X_n(0) Y_n(y) = 0$$

$$\sum_{n=0}^{\infty} [X_n''(x) - \lambda_n X_n(x)] Y_n(y) = \sum_{n=0}^{\infty} f_n Y_n(y)$$

$$u(a, y) = \sum_{n=0}^{\infty} X_n(a) Y_n(y) = Ay = \sum_{n=0}^{\infty} \psi_n Y_n(y)$$

建立初值定解问题

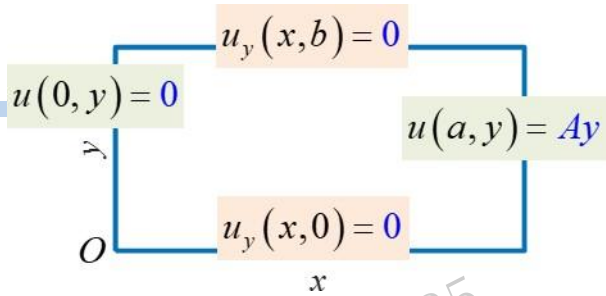
$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = f_n \\ X_n(0) = 0, \quad X_n(a) = \psi_n \end{cases}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = x, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(a, y) = Ay, & 0 \leq y \leq b \\ u_y(x, 0) = 0, \quad u_y(x, b) = 0, & 0 \leq x \leq a \end{cases}$$

Poisson/二维/非齐次		
x边界	1齐	1非齐
y边界	2齐	2齐



第4步 解x方向ODE

$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = f_n \\ X_n(0) = 0, \quad X_n(a) = \psi_n \end{cases}$$

$$\lambda_n = \left(\frac{n\pi}{b}\right)^2, \quad Y_n(x) = \cos\left(\frac{n\pi}{b}y\right), \quad (n = 0, 1, 2, 3, \dots)$$

$$f_n = \begin{cases} x, & (n=0) \\ 0, & (n>0) \end{cases}$$

$$\psi_n = \begin{cases} \frac{Ab}{2}, & (n=0) \\ \frac{2[(-1)^n - 1]}{n^2 \pi^2} Ab, & (n>0) \end{cases}$$

ODE类型随n不同而不同→分类讨论

① 当 $n=0$ 时

$$\lambda_0 = 0, \quad Y_0(x) = 1, \quad f_0 = x \neq 0$$

ODE可化简为

$$\begin{cases} X_0''(x) = f_0 = x \\ X_0(0) = 0, \quad X_0(a) = \psi_0 \end{cases} \quad \text{直接积分法}$$

通解为 $X_0(x) = \frac{x^3}{6} + \bar{C}_{n1}x + \bar{C}_{n2}$

根据边界条件

$$X_0(0) = \frac{x^3}{6} + \bar{C}_{n1}x + \bar{C}_{n2} = 0$$

$$X_0(a) = \frac{a^3}{6} + \bar{C}_{n1}a + \bar{C}_{n2} = \psi_0 = \frac{Ab}{2}$$

$$\begin{aligned} \bar{C}_{n1} &= \frac{Ab}{2a} - \frac{a^2}{6} \\ \bar{C}_{n2} &= 0 \end{aligned}$$

$n=0$ 时x方向ODE的解为

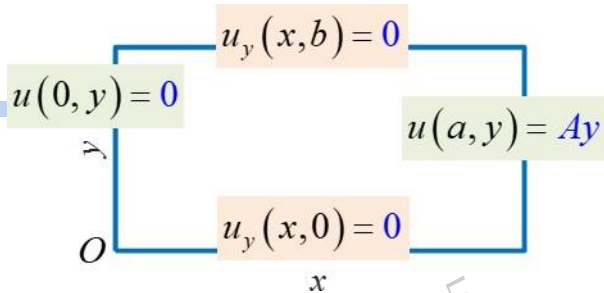
$$X_0(x) = \frac{x^3}{6} + \left(\frac{Ab}{2a} - \frac{a^2}{6}\right)x$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = x, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(a, y) = Ay, & 0 \leq y \leq b \\ u_y(x, 0) = 0, \quad u_y(x, b) = 0, & 0 \leq x \leq a \end{cases}$$

Poisson/二维/非齐次		
x边界	1齐	1非齐
y边界	2齐	2齐



第4步 解x方向ODE

$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = f_n \\ X_n(0) = 0, \quad X_n(a) = \psi_n \end{cases}$$

② 当 $n > 0$ 时 $\lambda_0 \neq 0, f_n = 0$

ODE可化简为

$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = 0 \\ X_n(0) = 0, \quad X_n(a) = \psi_n \end{cases}$$

二阶线性齐次ODE, 通解为

$$X_n(x) = \bar{C}_{n1} \sinh \frac{n\pi}{b} x + \bar{C}_{n2} \cosh \frac{n\pi}{b} x$$

$$\lambda_n = \left(\frac{n\pi}{b} \right)^2, \quad Y_n(y) = \cos \left(\frac{n\pi}{b} y \right), \quad (n = 0, 1, 2, 3, \dots)$$

$$f_n = \begin{cases} x, & (n=0) \\ 0, & (n>0) \end{cases} \quad \psi_n = \begin{cases} \frac{Ab}{2}, & (n=0) \\ \frac{2[(-1)^n - 1]}{n^2 \pi^2} Ab, & (n>0) \end{cases}$$

根据边界条件

$$X_n(0) = \bar{C}_{n1} \sinh \frac{n\pi}{b} x + \bar{C}_{n2} \cosh \frac{n\pi}{b} x = 0$$

$$X_n(a) = \bar{C}_{n1} \sinh \frac{n\pi}{b} a + \bar{C}_{n2} \cosh \frac{n\pi}{b} a = \psi_n = \frac{2[(-1)^n - 1]}{n^2 \pi^2} Ab$$

$$\bar{C}_{n1} = \frac{[(-1)^n - 1]}{n^2 \pi^2} \frac{2Ab}{\sinh \frac{n\pi a}{b}}, \quad \bar{C}_{n2} = 0$$

$n > 0$ 时x方向ODE的解为

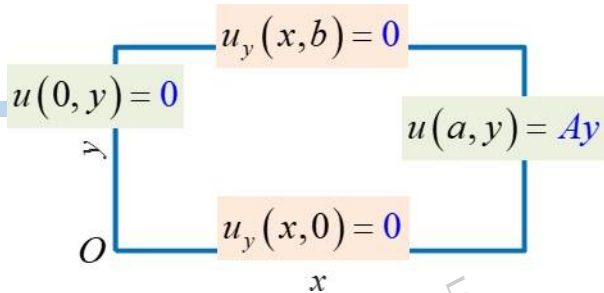
$$X_n(x) = \frac{2Ab[(-1)^n - 1]}{n^2 \pi^2} \cdot \frac{\sinh \frac{n\pi}{b} x}{\sinh \frac{n\pi}{b} a}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = x, & 0 < x < l, \quad 0 < y < h \\ u(0, y) = 0, \quad u(a, y) = Ay, & 0 \leq y \leq b \\ u_y(x, 0) = 0, \quad u_y(x, b) = 0, & 0 \leq x \leq a \end{cases}$$

Poisson/二维/非齐次		
x边界	1齐	1非齐
y边界	2齐	2齐

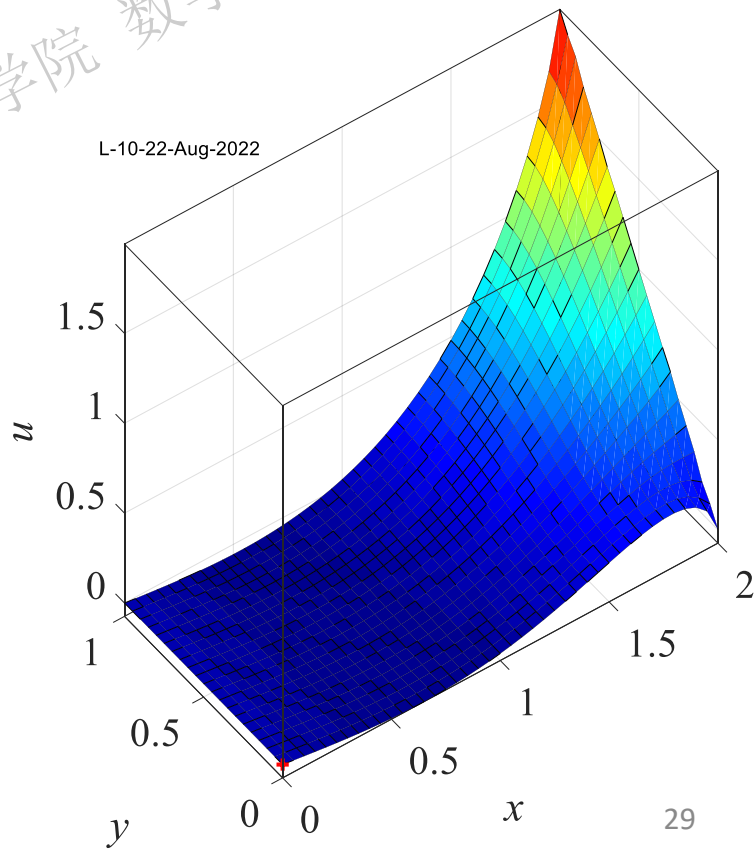


第5步 回代 原定解问题的解为

$$\lambda_n = \left(\frac{n\pi}{b} \right)^2, \quad Y_n(x) = \cos\left(\frac{n\pi}{b} y \right), \quad (n = 0, 1, 2, 3, \dots)$$

$$\begin{aligned} u(x, y) &= \sum_{n=0}^{\infty} X_n(x) Y_n(y) = X_0(x) \cdot 1 + \sum_{n=1}^{\infty} X_n(x) \cos\left(\frac{n\pi}{b} y \right) \\ &= \frac{x^3}{6} + \left(\frac{Ab}{2a} - \frac{a^2}{6} \right) x \\ &\quad + 2Ab \cdot \sum_{n=1}^{\infty} \frac{[(-1)^n - 1]}{n^2 \pi^2} \frac{\sinh \frac{n\pi}{b} x}{\sinh \frac{n\pi}{b} a} \cos\left(\frac{n\pi}{b} y \right) \end{aligned}$$

$$a=2, b=1, A=2$$





习题

例
$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次

x边界

y边界

解 第1步 坐标转换 环形区域→直角系无齐次边界→向极坐标转换

$$x = \rho \cos \theta, y = \rho \sin \theta$$

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = \frac{1}{2} \rho^2 \sin 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

第2步 特征值/特征函数

周向周期性边界

$$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$$

$$\lambda_n = n^2, \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, (n=1, 2, 3, \dots)$$

特征函数含有1, 需分类讨论

Laplace/二维/齐次

周向 θ

周期

周期

径向 ρ

1齐

1非齐

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3, \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3, \dots)$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho^2 \sin 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots)$$

第3步 正交分解 设原定解问题有形式解

$$u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta)$$

自由项按特征函数系展开

$$\frac{1}{2}\rho^2 \sin 2\theta = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

Fourier级数展开

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n=0, 1, 2, 3, \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$

$$n=0 \quad f_0 = \frac{1}{2} \cdot \frac{2}{2\pi} \int_0^{2\pi} \frac{1}{2} \rho \sin 2\theta \cdot 1 \cdot d\theta = \frac{\rho}{4\pi} \left[-\frac{1}{2} \cos 2\theta \right]_0^{2\pi} = 0$$

$$f_{n1} = \frac{2}{2\pi} \int_0^{2\pi} \frac{1}{2} \rho \sin 2\theta \cdot \cos n\theta \cdot d\theta = \begin{cases} \frac{\rho}{2\pi} \left[-\frac{\cos 4\theta}{8} \right]_0^{2\pi} = 0, & (n=2) \\ \frac{\rho}{2\pi} \left[-\frac{\cos(2+n)\theta}{2(2+n)} - \frac{\cos(2-n)\theta}{2(2-n)} \right]_0^{2\pi} = 0, & (n \neq 2) \end{cases}$$

$$f_{n2} = \frac{2}{2\pi} \int_0^{2\pi} \frac{1}{2} \rho \sin 2\theta \cdot \sin n\theta \cdot d\theta = \begin{cases} \frac{\rho}{2\pi} \left[\frac{1}{4} (2\theta - \sin 2\theta \cos 2\theta) \right]_0^{2\pi} = \frac{\rho}{2}, & (n=2) \\ \frac{\rho}{2\pi} \left[-\frac{\sin(2+n)\theta}{2(2+n)} + \frac{\sin(2-n)\theta}{2(2-n)} \right]_0^{2\pi} = 0, & (n \neq 2) \end{cases}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次

x边界

y边界

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次

周向 θ

周期

周期

径向 ρ

1齐

1非齐

第3步 正交分解 $u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta)$

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots)$$

Fourier级数展开

径向外边界按照特征函数系展开

$$a(\cos \theta + \sin \theta) = \sum_{n=0}^{\infty} \varphi_n \Phi_n(\theta)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n=0, 1, 2, 3, \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$

$$\begin{cases} n=0 \\ \varphi_0 = \frac{1}{2} \cdot \frac{2}{2\pi} \int_0^{2\pi} a(\cos \theta + \sin \theta) \cdot 1 \cdot d\theta = \frac{a}{2\pi} [\sin \theta - \cos \theta]_0^{2\pi} = 0 \\ n>0 \\ \varphi_{n1} = \frac{2}{2\pi} \int_0^{2\pi} a(\cos \theta + \sin \theta) \cdot \cos n\theta \cdot d\theta \end{cases}$$

$$= \frac{a}{\pi} \left[\int_0^{2\pi} \cos \theta \cdot \cos n\theta \cdot d\theta + \int_0^{2\pi} \sin \theta \cdot \cos n\theta \cdot d\theta \right]$$

$$= \begin{cases} \frac{a}{\pi} \left[\frac{1}{2} (\theta + \sin \theta \cos \theta) - \frac{1}{4} \cos 2\theta \right]_0^{2\pi} = a, & (n=1) \\ \frac{a}{\pi} \left[\frac{\sin(1+n)\theta}{2(1+n)} + \frac{\sin(1-n)\theta}{2(1-n)} - \frac{\cos(1+n)\theta}{2(1+n)} - \frac{\cos(1-n)\theta}{2(1-n)} \right]_0^{2\pi} = 0, & (n \neq 1) \end{cases}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\begin{aligned} \lambda_0 &= 0, \quad \Phi_0(\theta) = 1, \quad (n=0) \\ \lambda_n &= n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots) \end{aligned}$$

第3步 正交分解 $u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta)$

径向外边界按照特征函数系展开

$$a(\cos \theta + \sin \theta) = \sum_{n=0}^{\infty} \varphi_n \Phi_n(\theta)$$

$$\begin{aligned} n > 0 \quad \varphi_{n2} &= \frac{2}{2\pi} \int_0^{2\pi} a(\cos \theta + \sin \theta) \cdot \sin n\theta \cdot d\theta \\ &= \frac{a}{\pi} \left[\int_0^{2\pi} \cos \theta \cdot \sin n\theta \cdot d\theta + \int_0^{2\pi} \sin \theta \cdot \sin n\theta \cdot d\theta \right] \\ &= \begin{cases} \frac{a}{\pi} \left[-\frac{1}{4} \cos 2\theta + \frac{1}{2}(\theta - \sin \theta \cos \theta) \right]_0^{2\pi} = a, & (n=1) \\ \frac{a}{\pi} \left[-\frac{\cos(1+n)\theta}{2(1+n)} + \frac{\cos(1-n)\theta}{2(1-n)} - \frac{\sin(1+n)\theta}{2(1+n)} + \frac{\sin(1-n)\theta}{2(1-n)} \right]_0^{2\pi} = 0, & (n \neq 1) \end{cases} \end{aligned}$$

Fourier级数展开

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \\ a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n=0, 1, 2, 3, \dots) \\ b_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots) \end{aligned}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots)$$

第3步 正交分解 $u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta)$

汇总

自由项

$$\frac{1}{2}\rho \sin 2\theta = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

$$f_n = \begin{cases} (n=0), & 0, \quad (\Phi_0 = 1) \\ (n>0), & \begin{cases} f_{n1} = 0, & (\Phi_n = \cos n\theta) \\ f_{n2} = \begin{cases} \frac{\rho}{2}, & (n=2) \\ 0, & (n \neq 2) \end{cases}, & (\Phi_n = \sin n\theta) \end{cases} \end{cases}$$

径向外边界

$$a(\cos \theta + \sin \theta) = \sum_{n=0}^{\infty} \varphi_n \Phi_n(\theta)$$

$$\varphi_n = \begin{cases} (n=0), & 0, \quad (\Phi_0 = 1) \\ (n>0), & \begin{cases} \varphi_{n1} = \begin{cases} a, & (n=1) \\ 0, & (n \neq 1) \end{cases}, & (\Phi_n = \cos n\theta) \\ \varphi_{n2} = \begin{cases} a, & (n=1) \\ 0, & (n \neq 1) \end{cases}, & (\Phi_n = \sin n\theta) \end{cases} \end{cases}$$

后续径向ODE的求解需分段讨论

$$n=0, \quad n=1, \quad n=2, \quad n \geq 3$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots)$$

第4步 建立径向ODE

设原定解问题有形式解 $u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta)$

$$\sum_{n=0}^{\infty} R_n''(\rho) \Phi_n(\theta) + \frac{1}{\rho} \sum_{n=0}^{\infty} R_n'(\rho) \Phi_n(\theta) + \frac{1}{\rho^2} \sum_{n=0}^{\infty} R_n(\rho) \Phi_n''(\theta) = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

$$\sum_{n=0}^{\infty} R_n''(\rho) \Phi_n(\theta) + \sum_{n=0}^{\infty} \frac{1}{\rho} R_n'(\rho) \Phi_n(\theta) + \sum_{n=0}^{\infty} \frac{1}{\rho^2} R_n(\rho) [-\lambda_n \Phi_n(\theta)] = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

$$\sum_{n=0}^{\infty} \left[R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) + \frac{-\lambda_n}{\rho^2} R_n(\rho) \right] \Phi_n(\theta) = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

$$\begin{cases} \rho^2 R_n''(\rho) + \rho R_n'(\rho) - \lambda_n R_n(\rho) = f_n \\ R_n(b) = 0, R_n(a) = \varphi_n \end{cases}$$

径向边界条件 $u(b, \theta) = \sum_{n=0}^{\infty} R_n(b) \Phi_n(\theta) = 0$

条件 $u(a, \theta) = \sum_{n=0}^{\infty} R_n(a) \Phi_n(\theta) = a(\cos \theta + \sin \theta) = \sum_{n=0}^{\infty} \varphi_n \Phi_n(\theta)$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3 \dots)$$

第5步 解径向ODE

$$\begin{cases} \rho^2 R_n''(\rho) + \rho R_n'(\rho) - \lambda_n R_n(\rho) = f_n \\ R_n(b) = 0, R_n(a) = \varphi_n \end{cases}$$

① 当 $n=0$ 时 $\lambda_0 = 0, f_0 = 0, \varphi_0 = 0$

$$\begin{cases} \rho^2 R_0''(\rho) + \rho R_0'(\rho) = 0 \\ R_0(b) = 0, R_0(a) = 0 \end{cases}$$

齐次Euler方程, 通解为 $R_0(\rho) = C_{01} \ln \rho + C_{02}$

由径向边界条件

$$R_0(b) = C_{01} \ln b + C_{02} = 0$$

$$R_0(a) = C_{01} \ln a + C_{02} = 0$$

$$C_{01} = C_{02} = 0$$

Euler 方程的解

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0$$

$$\begin{cases} a_n < 0, & x = C_1 t^{\sqrt{-a_n}} + C_2 t^{-\sqrt{-a_n}} \\ a_n = 0, & x = C_1 \ln t + C_2 \\ a_n > 0, & x = C_1 \cos(\sqrt{a_n} \ln t) + C_2 \sin(\sqrt{a_n} \ln t) \end{cases}$$

$n=0$ 时径向ODE解为 $R_0(\rho) = 0$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots)$$

第5步 解径向ODE

$$\begin{cases} \rho^2 R_n''(\rho) + \rho R_n'(\rho) - \lambda_n R_n(\rho) = f_n \\ R_n(b) = 0, R_n(a) = \varphi_n \end{cases}$$

② 当 $n=1$ 时 $\lambda_1 = 1, f_1 = 0, \varphi_{1,1} = \varphi_{1,2} = a$

$$\begin{cases} \rho^2 R_1''(\rho) + \rho R_1'(\rho) - \lambda_1 R_1(\rho) = 0 \\ R_1(b) = 0, R_1(a) = \varphi_{1,1} = \varphi_{1,2} = a \end{cases}$$

齐次Euler方程, 通解为 $R_1(\rho) = C_{11}\rho + C_{12}\rho^{-1}$

由径向边界条件

$$\begin{aligned} R_1(b) &= C_{11}b + C_{12}b^{-1} = 0 \\ R_1(a) &= C_{11}a + C_{12}a^{-1} = a \end{aligned} \quad \begin{aligned} C_{11} &= \frac{a^2}{a^2 - b^2} \\ C_{12} &= -\frac{a^2 b^2}{a^2 - b^2} \end{aligned}$$

Euler方程的解

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0$$

$$\begin{cases} a_n < 0, & x = C_1 t^{\sqrt{-a_n}} + C_2 t^{-\sqrt{-a_n}} \\ a_n = 0, & x = C_1 \ln t + C_2 \\ a_n > 0, & x = C_1 \cos(\sqrt{a_n} \ln t) + C_2 \sin(\sqrt{a_n} \ln t) \end{cases}$$

$n=1$ 时径向ODE解为

$$R_1(\rho) = \frac{a^2}{a^2 - b^2} \rho - \frac{a^2 b^2}{a^2 - b^2} \rho^{-1}$$

该解同时适用于特征函数

$$\Phi_{1,1}(\theta) = \cos \theta, \quad \Phi_{1,2}(\theta) = \sin \theta$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3 \dots)$$

第5步 解径向ODE

$$\begin{cases} \rho^2 R_n''(\rho) + \rho R_n'(\rho) - \lambda_n R_n(\rho) = f_n \\ R_n(b) = 0, R_n(a) = \varphi_n \end{cases}$$

③ 当 $n=2, \Phi_2(\theta) = \cos 2\theta$ 时

$$\lambda_2 = 2^2, \quad f_{2,1} = 0, \quad \varphi_{2,1} = \varphi_{2,2} = 0$$

$$\begin{cases} \rho^2 R_2''(\rho) + \rho R_2'(\rho) - \lambda_2 R_2(\rho) = f_{2,1} = 0 \\ R_2(b) = 0, R_2(a) = 0 \end{cases}$$

齐次Euler方程, 通解为 $R_2(\rho) = C_{21}t^2 + C_{22}t^{-2}$

根据径向边界条件

$$\begin{aligned} R_2(b) &= C_{21}b^2 + C_{22}b^{-2} = 0 \\ R_2(a) &= C_{21}a^2 + C_{22}a^{-2} = 0 \end{aligned} \quad C_{21} = C_{22} = 0$$

Euler 方程的解

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0$$

$$\begin{cases} a_n < 0, & x = C_1 t^{\sqrt{-a_n}} + C_2 t^{-\sqrt{-a_n}} \\ a_n = 0, & x = C_1 \ln t + C_2 \\ a_n > 0, & x = C_1 \cos(\sqrt{a_n} \ln t) + C_2 \sin(\sqrt{a_n} \ln t) \end{cases}$$

$n=2$ 径向ODE的解为

$$R_2(\rho) = 0$$

该解仅适于特征函数 $\Phi_{2,1}(\theta) = \cos 2\theta$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

第5步 解径向ODE

$$\begin{cases} \rho^2 R_n''(\rho) + \rho R_n'(\rho) - \lambda_n R_n(\rho) = f_n \\ R_n(b) = 0, R_n(a) = \varphi_n \end{cases}$$

③ 当 $n=2, \Phi_2(\theta) = \sin 2\theta$ 时

$$\lambda_2 = 2^2, f_2 = \frac{\rho}{2}, \varphi_{2,1} = \varphi_{2,2} = 0$$

$$\begin{cases} \rho^2 R_2''(\rho) + \rho R_2'(\rho) - \lambda_2 R_2(\rho) = f_{2,2} = \frac{\rho}{2} \\ R_2(b) = 0, R_2(a) = 0 \end{cases}$$

非齐次Euler方程 令 $\rho = \exp s$, 则 $s = \ln \rho$

$$R'(\rho) = \frac{dR}{ds} \frac{ds}{d\rho} = \frac{1}{\rho} \frac{dR}{ds} \quad R''(\rho) = -\frac{1}{\rho^2} \frac{dR}{ds} + \frac{1}{\rho^2} \frac{d^2 R}{ds^2}$$

$$\rho^2 \left(-\frac{1}{\rho^2} \frac{dR}{ds} + \frac{1}{\rho^2} \frac{d^2 R}{ds^2} \right) + \rho \left(\frac{1}{\rho} \frac{dR}{ds} \right) - \lambda_2 R_2(\rho) = \frac{\exp s}{2}$$

$$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$$

$$\lambda_n = n^2, \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, (n=1, 2, 3 \dots)$$

$$\frac{d^2 R_2(s)}{ds^2} - \lambda_2 R_2(s) = \frac{1}{2} \exp s$$

$$R(s) = C_{21} \exp(2s) + C_{21} \exp(-2s) + \bar{R}$$

特解形式为 $\bar{R} = \bar{C}_1 \exp s$

$$\text{各阶导数} \quad \bar{R}' = \bar{R}'' = \bar{C}_1 \exp s$$

$$\text{代入ODE} \quad \bar{C}_1 \exp s - \lambda_2 \bar{C}_1 \exp s = \frac{1}{2} \exp s$$

$$\text{得} \quad \bar{C}_1 = \frac{1}{2(1-\lambda_2)} = -\frac{1}{6}$$

$$\text{特解为} \quad \bar{R} = -\frac{1}{6} \exp s = -\frac{\rho}{6}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, \quad 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3 \dots)$$

第5步 解径向ODE

③ 当 $n=2, \Phi_2(\theta) = \sin 2\theta$ 时

$$\begin{cases} \rho^2 R_2''(\rho) + \rho R_2'(\rho) - \lambda_2 R_2(\rho) = f_2 = \frac{\rho}{2} \\ R_2(b) = 0, R_2(a) = 0 \end{cases}$$

非齐次Euler方程通解为

$$R(s) = C_{21} \exp(2s) + C_{22} \exp(-2s) - \frac{1}{6} \exp s$$

还原变量 $\rho = \exp s, s = \ln \rho$

$$R(\rho) = C_{21} \rho^2 + C_{22} \rho^{-2} - \frac{\rho}{6}$$

根据径向边界条件

$$\begin{aligned} R(b) &= C_{21} b^2 + C_{22} b^{-2} - \frac{b}{6} = 0 \\ R(a) &= C_{21} a^2 + C_{22} a^{-2} - \frac{a}{6} = 0 \end{aligned}$$

解得

$$C_{21} = \frac{1}{6} \left(\frac{a^3 - b^3}{a^4 - b^4} \right), \quad C_{22} = \frac{a^3 b^3}{6} \left[\frac{a - b}{a^4 - b^4} \right]$$

$n=2, \Phi_2(\theta) = \sin 2\theta$ 时, 径向ODE通解为

$$R_2(\rho) = \frac{1}{6} \left(\frac{a^3 - b^3}{a^4 - b^4} \right) \rho^2 + \frac{a^3 b^3}{6} \left[\frac{a - b}{a^4 - b^4} \right] \rho^{-2} - \frac{\rho}{6}$$

习题

例

$$\begin{cases} u_{xx} + u_{yy} = xy, & b^2 < x^2 + y^2 < a^2 \\ u(x, y) = 0, & x^2 + y^2 = b^2 \\ u(x, 0) = x + y, & x^2 + y^2 = a^2 \end{cases}$$

Poisson/二维/非齐次		
x边界		
y边界		

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\theta\theta} = \frac{1}{2}\rho \sin 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(b, \theta) = 0, & \rho = b \\ u(a, \theta) = a(\cos \theta + \sin \theta), & \rho = a \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	周期
径向 ρ	1齐	1非齐

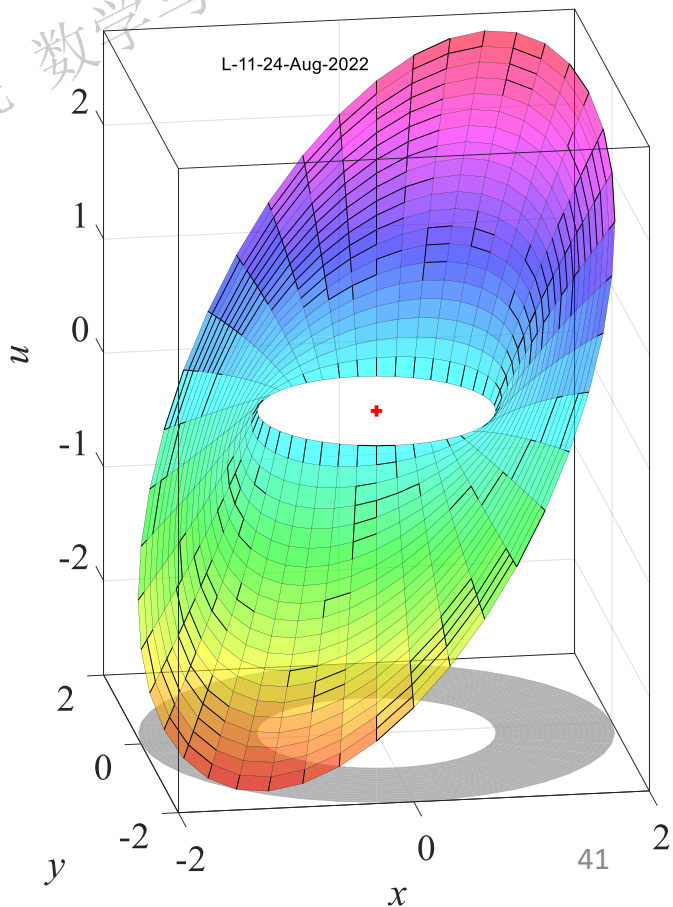
$$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, \quad (n=0)$$

$$\lambda_n = n^2, \quad \Phi_n(\theta) = C_{n1} \cos n\theta + C_{n2} \sin n\theta, \quad (n=1, 2, 3, \dots)$$

第6步 回代

原定解问题的解可表示为

$$\begin{aligned} u(\rho, \theta) &= \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta) \\ &= R_0(\rho) \Phi_0(\theta) + R_1(\rho) \Phi_1(\theta) + R_2(\rho) \Phi_2(\theta) + \sum_{n=3}^{\infty} R_n(\rho) \Phi_n(\theta) \\ &= \left(\frac{a^2}{a^2 - b^2} \rho - \frac{a^2 b^2}{a^2 - b^2} \rho^{-1} \right) (\cos \theta + \sin \theta) \\ &\quad + \left[\frac{\rho^2}{6} \cdot \frac{a^3 - b^3}{a^4 - b^4} + \frac{a^3 b^3}{6 \rho^2} \cdot \frac{a - b}{a^4 - b^4} - \frac{\rho}{6} \right] \sin 2\theta \end{aligned}$$





习题

例

$$\begin{cases} u_t + (1+x^2)u_x - u = 0, & 0 < t, \quad -\infty < x < +\infty \\ u(x, 0) = \arctan x, & -\infty < x < +\infty \end{cases}$$

解 原PDE是无界Cauchy问题, 可用特征线法

第1步 求特征线方程

根据原PDE, 特征线应满足 $\frac{dx}{dt} = 1 + x^2$

$$\arctan x = t + \tau$$

解得特征线方程为 $x = \tan(t + \tau)$

第2步 改造原PDE/定解条件

原PDE可表示为 $u_t + (1+x^2)u_x - u = 0$

$$u_t + u_x \frac{dx}{dt} - u = 0$$

原初始条件可表示为 $\frac{du}{dt} - u = 0$

$$u(x, 0) = \arctan x = \arctan[\tan(t + \tau)] = \tau$$

原定解问题可转化为ODE

$$\begin{cases} \frac{du}{dt} - u = 0, & 0 < t, \quad -\infty < x < +\infty \\ u(0) = \tau, & -\infty < x < +\infty \end{cases}$$

第3步 解ODE定解问题

分离变量法 $u(t) = C \exp t$

根据初始条件 $u(0) = C = \tau$

ODE的解 $u(t) = \tau \exp t$

第4步 消特征线常数 由特征线方程知

$\tau = \arctan x - t$ 代入上述解

得原PDE定解问题的解

$$u(x, t) = (\arctan x - t) \exp t$$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = x, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

2齐

2齐

解 第1步 确定特征值/特征函数

边界已经齐次, [2, 2]型边界

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x \right), (n = 0, 1, 2, 3 \dots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n = 1, 2, 3 \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x \right), (n = 0, 1, 2, 3 \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n = 0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n = 1, 2, 3 \dots)$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = x, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

2齐

2齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x \right), (n = 0, 1, 2, 3, \dots)$$

第2步 正交分解

设原问题有分离变量的形式解

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos\left(\frac{n\pi}{l} x \right)$$

自由项/初始条件按特征函数系展开

$$u(x, 0) = x = \sum_{n=0}^{\infty} \varphi_n X_n(x) = \sum_{n=0}^{\infty} \varphi_n \cos\left(\frac{n\pi}{l} x \right)$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$n=0 \quad \varphi_0 = \frac{1}{l} \int_0^l x \cdot 1 dx = \frac{l}{2}$$

$$n>0 \quad \varphi_n = \frac{2}{l} \int_0^l x \cos \frac{n\pi x}{l} \cdot dx = \frac{2}{l} \left[\frac{l}{n\pi} x \sin \frac{n\pi}{l} x + \frac{l^2}{n^2 \pi^2} \cos \frac{n\pi}{l} x \right]_0^l = \frac{2l}{n^2 \pi^2} [(-1)^n - 1]$$

不能兼容 $n=0$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = x, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

2齐

2齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n = 0, 1, 2, 3, \dots)$$

第3步 建立初值问题ODE

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos\left(\frac{n\pi}{l}x\right)$$

假设形式解可逐项求导, 代入原PDE

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) X_n''(x) = 0$$

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) [-\lambda_n X_n(x)] = 0$$

$$\sum_{n=0}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = 0$$

$$T_n''(t) + a^2 \lambda_n T_n(t) = 0$$

用原PDE初始条件→初值问题ODE初始条件

$$u(x, 0) = \sum_{n=0}^{\infty} T_n(0) X_n(x) = x = \sum_{n=0}^{\infty} \varphi_n X_n(x)$$

$$u_t(x, 0) = \sum_{n=0}^{\infty} T_n'(0) X_n(x) = 0 = \sum_{n=0}^{\infty} \psi_n X_n(x)$$

建立了初值问题ODE

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n, T_n'(0) = 0 \end{cases}$$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = x, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

2齐

2齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n = 0, 1, 2, 3, \dots)$$

第4步 求初值问题

二阶线性
齐次ODE

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n, T_n'(0) = 0 \end{cases}$$

① 当 $n=0$ 时

$$T_0(t) = C_{1,0} \cdot t + C_{2,0}$$

$$T_0'(t) = C_{1,0}$$

$$\begin{cases} T_0(0) = C_{2,0} = \varphi_0 \\ T_0'(0) = C_{1,0} = 0 \end{cases} \rightarrow \begin{cases} C_{1,0} = 0 \\ C_{2,0} = \varphi_0 = \frac{l}{2} \end{cases}$$

初值问题的解

$$T_0(t) = \frac{l}{2}$$

② 当 $n>0$ 时

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t)$$

$$T_n'(t) = -C_{1n}a\sqrt{\lambda_n} \sin(a\sqrt{\lambda_n}t) + C_{2n}a\sqrt{\lambda_n} \cos(a\sqrt{\lambda_n}t)$$

$$\begin{cases} T_n(0) = C_{1n} = \varphi_n \\ T_n'(0) = C_{2n}a\sqrt{\lambda_n} = 0 \end{cases} \rightarrow \begin{cases} C_{1n} = \varphi_n \\ C_{2n} = 0 \end{cases}$$

初值问题的解

$$T_n(t) = \varphi_n \cos\left(\frac{n\pi a}{l}t\right)$$



习题

例

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) = 0, & t \geq 0 \\ u(x, 0) = x, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

2齐

2齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n = 0, 1, 2, 3, \dots)$$

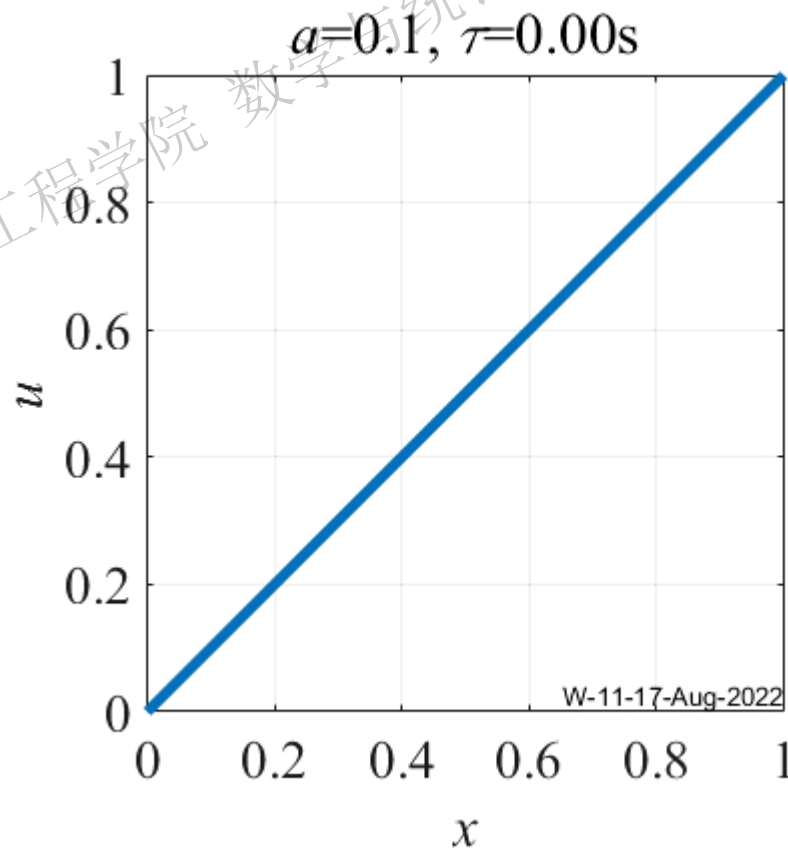
第4步 求初值问题

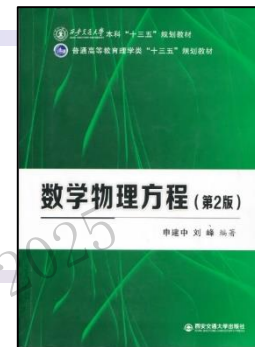
原定解问题的解可表示为

$$\begin{aligned} u(x, t) &= \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos\left(\frac{n\pi}{l}x\right) \\ &= \frac{l}{2} + \sum_{n=1}^{\infty} \frac{2l}{n^2\pi^2} [(-1)^n - 1] \cos\left(\frac{n\pi a}{l}t\right) \cos\left(\frac{n\pi}{l}x\right) \end{aligned}$$

单列

特征函数有1时→其平方模与sin/cos
不同→正交分解需单独讨论n=0





习题

例

$$\begin{cases} u_{tt} - u_{xx} - 4u = 2\sin^2 x, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 0, u_x(\pi, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

波动/一维/非齐

2齐

2齐

解 边界已经齐次, $[2, 2]$ 型边界 $\lambda_n = \left(\frac{n\pi}{l}\right)^2$, $X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$, ($n = 0, 1, 2, 3, \dots$)

本例中 $l = \pi$, $\lambda_n = n^2$, $X_n(x) = \cos nx$, ($n = 0, 1, 2, 3, \dots$)

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

正交分解, 原问题有分离变量的形式解

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos nx$$

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

自由项/初始条件按特征函数系展开

$$2\sin^2 x = \sum_{n=0}^{\infty} f_n X_n(x) = \sum_{n=0}^{\infty} f_n \cos nx$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\left\{ \begin{array}{l} \textcircled{1} \quad n=0 \quad f_0 = \frac{2}{2\pi} \int_0^\pi 2\sin^2 x \cdot 1 \cdot dx = \frac{1}{\pi} \int_0^\pi (1 - \cos 2x) \cdot 1 \cdot dx = 1 \end{array} \right.$$

$$\left\{ \begin{array}{l} \textcircled{2} \quad n>0 \quad f_n = \frac{2}{l} \int_0^l 2\sin^2 x \cos nx dx = \frac{4}{\pi} \int_0^\pi \sin^2 x \cos nx dx = \begin{cases} -1 & (n=2) \\ 0 & (n>2) \end{cases} \end{array} \right.$$

习题

例

$$\begin{cases} u_{tt} - u_{xx} - 4u = 2\sin^2 x, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 0, u_x(\pi, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

波动/一维/非齐

2齐

2齐

$$l = \pi, \lambda_n = n^2, X_n(x) = \cos nx, (n = 0, 1, 2, 3, \dots)$$

建立初值ODE, 将正交分解后的形式解/自由项代入原PDE

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) - \sum_{n=0}^{\infty} T_n(t) X_n''(x) - 4 \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} f_n X_n(x)$$

根据原初始条件

$$\sum_{n=0}^{\infty} T_n''(t) X_n(x) + \sum_{n=0}^{\infty} T_n(t) \lambda_n X_n(x) - 4 \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} f_n X_n(x)$$

$$u(x, 0) = \sum_{n=0}^{\infty} T_n(0) X_n(x) = 0$$

$$\sum_{n=0}^{\infty} [T_n''(t) + (\lambda_n - 4)T_n(t)] X_n(x) = \sum_{n=0}^{\infty} f_n X_n(x)$$

$$u_t(x, 0) = \sum_{n=0}^{\infty} T_n'(0) X_n(x) = 0$$

n 的取值
影响方程

初值ODE
定解问题

$$\begin{cases} T_n''(t) + (n^2 - 4)T_n(t) = f_n \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

$$f_n = \begin{cases} 1 & (n=0) \\ -1 & (n=2) \\ 0 & (n \neq 0, 2) \end{cases}$$

n 的取值影响方程性态+自由项

习题

例

$$\begin{cases} u_{tt} - u_{xx} - 4u = 2\sin^2 x, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 0, u_x(\pi, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

波动/一维/非齐
2齐 2齐

$$l = \pi, \lambda_n = n^2, X_n(x) = \cos nx, (n=0, 1, 2, 3 \dots)$$

求解初值ODE

$$\begin{cases} T_n''(t) + (n^2 - 4)T_n(t) = f_n \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

特征方程定通解

$$s^2 + (n^2 - 4) = 0$$

自由项
定特解

$$f_n = \begin{cases} 1 & (n=0) \\ -1 & (n=2) \\ 0 & (n \neq 0, 2) \end{cases}$$

$n=0$

$$\begin{cases} T_0''(t) - 4T_0(t) = 1 \\ T_0(0) = 0, T_0'(0) = 0 \end{cases}$$

$$T_0(t) = C_{01}e^{2t} + C_{02}e^{-2t} + \bar{T}_0(t)$$

$$T_0(t) = \frac{1}{8}(e^{2t} + e^{-2t}) - \frac{1}{4}$$

$n=1$

$$\begin{cases} T_1''(t) - 3T_1(t) = 0 \\ T_1(0) = 0, T_1'(0) = 0 \end{cases}$$

$$T_1(t) = C_{01}e^{\sqrt{3}t} + C_{02}e^{-\sqrt{3}t}$$

$$T_1(t) = 0$$

$n=2$

$$\begin{cases} T_2''(t) = -1 \\ T_2(0) = 0, T_2'(0) = 0 \end{cases}$$

$$T_2(t) = -\frac{t^2}{2} + C_{21}t + C_{02}$$

$$T_2(t) = -\frac{t^2}{2}$$

$n \geq 3$

$$\begin{cases} T_n''(t) + (n^2 - 4)T_n(t) = 0 \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

$$T_n(t) = C_{n1} \cos \sqrt{n^2 - 4}t + C_{n2} \sin \sqrt{n^2 - 4}t$$

$$T_n(t) = 0$$

本质: 根据叠加原理对自由项的拆分

$$u(x, t) = T_0(t)X_0(x) + T_1(t)X_1(x) + T_2(t)X_2(x) + \sum_{n=3}^{\infty} T_n(t)X_n(x)$$

$$= \frac{1}{8}(e^{2t} + e^{-2t}) - \frac{1}{4} - \frac{1}{2}t^2 \cos 2x$$

习题

例

$$\begin{cases} u_{tt} - u_{xx} - 4u = 2\sin^2 x, & 0 < x < \pi, t > 0 \\ u_x(0, t) = 0, u_x(\pi, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq \pi \end{cases}$$

波动/一维/非齐

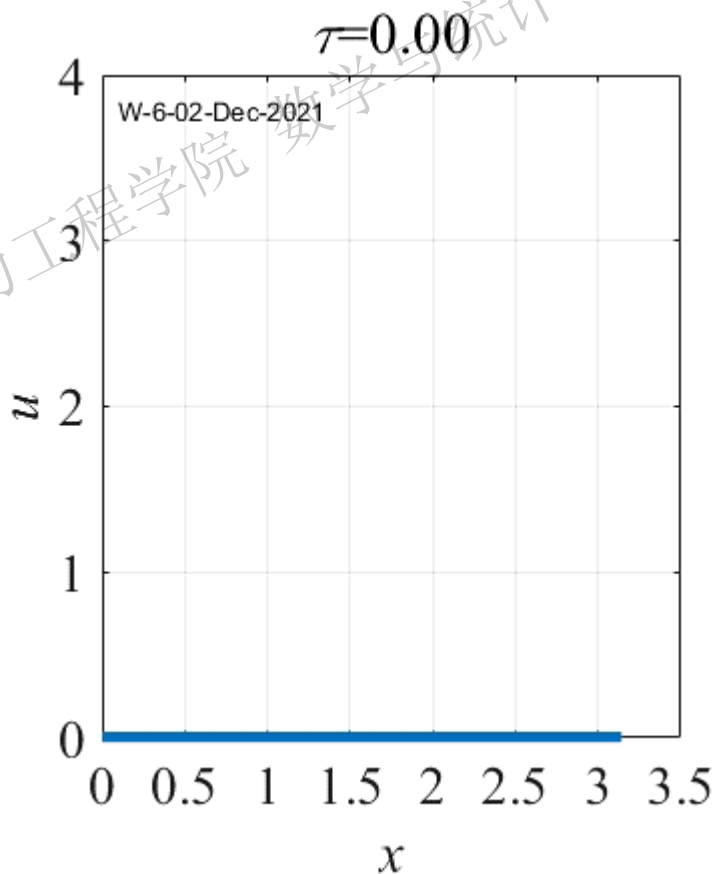
2齐

2齐

$$l = \pi, \lambda_n = n^2, X_n(x) = \cos nx, (n = 0, 1, 2, 3 \dots)$$

原定解问题的解

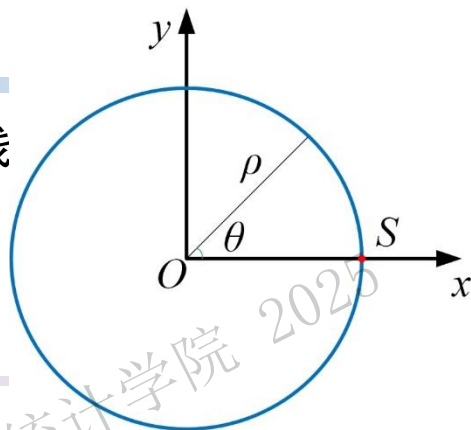
$$u(x, t) = \frac{1}{8}(e^{2t} + e^{-2t}) - \frac{1}{4} - \frac{1}{2}t^2 \cos 2x$$



习题

例

如图所示, 细金属丝被弯成半径为 ρ 的圆环, 经焊接制成环形线圈. 取焊接结束为0时刻, 此时圆环内的温度分布为 $A \cos^2 \frac{\theta}{2}$. 请写出描述该环形线圈温度演化的定解问题并求解



解

导热现象/无内热源

$$u_t = a^2 \Delta u$$

$$u_t = a^2 \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} \right)$$

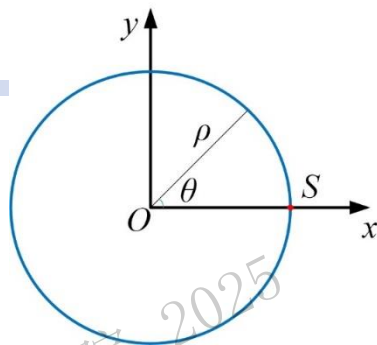
$$\Delta u = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}$$

极坐标下展开 $\Delta u \rightarrow$ 径向导数项为0

↓ 周期性边界

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}, & 0 \leq \theta < 2\pi, t > 0 \\ u(0, t) = u(2\pi, t), u_\theta(0, t) = u_\theta(2\pi, t), & 0 \leq t \\ u(\theta, 0) = \varphi(\theta) = A \cos^2 \frac{\theta}{2}, & 0 \leq \theta < 2\pi \end{cases}$$

习题



例

如图所示, 细金属丝被弯成半径为 ρ 的圆环, 经焊接制成环形线圈. 取焊接结束为0时刻, 此时圆环内的温度分布为 $A \cos^2 \frac{\theta}{2}$.

请写出描述该环形线圈温度演化的定解问题并求解

第1步 特征值问题 设 $u(\theta, t) = \Phi(\theta)T(t)$

代入原PDE有 $\Phi(\theta)T'(t) = \frac{a}{\rho^2} \Phi''(\theta)T(t)$

分离变量 $\frac{T'(t)}{\frac{a^2}{\rho^2} T(t)} = \frac{\Phi''(\theta)}{\Phi(\theta)} = -\lambda$

初值问题

$$T'(t) + \lambda \frac{a^2}{\rho^2} T(t) = 0 \quad \left\{ \begin{array}{l} \Phi''(\theta) + \lambda \Phi(\theta) = 0, \quad 0 \leq \theta \leq 2\pi \\ \Phi(0) = \Phi(2\pi), \quad \Phi'(0) = \Phi'(2\pi) \end{array} \right.$$

边值问题

特征值
特征函数

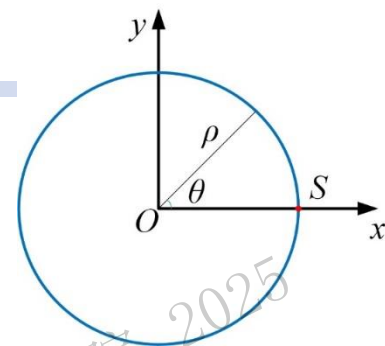
$$\lambda_n = n^2, \quad \Phi_n(\theta) = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}, & 0 \leq \theta < 2\pi, \quad t > 0 \\ u(0, t) = u(2\pi, t), \quad u_\theta(0, t) = u_\theta(2\pi, t), & 0 \leq t \\ u(\theta, 0) = A \cos^2 \frac{\theta}{2}, & 0 \leq \theta < 2\pi \end{cases}$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, \quad X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, \quad X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3, \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \quad \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \quad \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3, \dots)$

习题



例

如图所示, 细金属丝被弯成半径为 ρ 的圆环, 经焊接制成环形线圈. 取焊接结束为0时刻, 此时圆环内的温度分布为 $A \cos^2 \frac{\theta}{2}$.

请写出描述该环形线圈温度演化的定解问题并求解

第2步 正交分解 原PDE有形式解 $u(\theta, t) = \sum_{n=0}^{\infty} T_n(t) \Phi_n(\theta)$

将初始条件正交分解

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\varphi(\theta) = \sum_{n=0}^{\infty} \varphi_n \cdot \Phi_n(\theta) = A \cos^2 \frac{\theta}{2} = \frac{A}{2} \cos \theta + \frac{A}{2}$$

$$\varphi_0 = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{A}{2} \cos \theta + \frac{A}{2} \right) \cdot 1 \cdot d\theta = \frac{1}{2\pi} \cdot \frac{A}{2} \int_0^{2\pi} 1 \cdot d\theta = \frac{A}{2}$$

$$\varphi_{n,\cos} = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{A}{2} \cos \theta + \frac{A}{2} \right) \cdot \cos n\theta \cdot d\theta$$

$$= \frac{1}{\pi} \int_0^{2\pi} \left(\frac{A}{2} \cos \theta \right) \cdot \cos n\theta \cdot d\theta + \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{A}{2} \right) \cdot \cos n\theta \cdot d\theta = \frac{1}{\pi} \frac{A}{2} \int_0^{2\pi} \cos \theta \cdot \cos n\theta \cdot d\theta = \begin{cases} \frac{A}{2}, & (n=1) \\ 0, & (n \neq 1) \end{cases}$$

$$\varphi_{n,\sin} = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{A}{2} \cos \theta + \frac{A}{2} \right) \cdot \sin n\theta \cdot d\theta$$

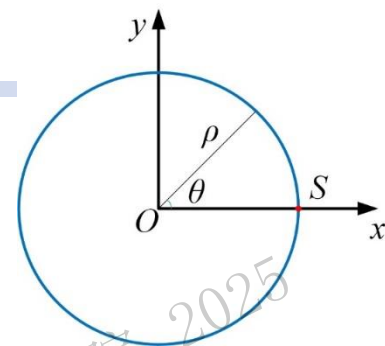
$$= \frac{1}{\pi} \int_0^{2\pi} \left(\frac{A}{2} \cos \theta \right) \cdot \sin n\theta \cdot d\theta + \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{A}{2} \right) \cdot \sin n\theta \cdot d\theta = 0$$

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}, & 0 \leq \theta < 2\pi, t > 0 \\ u(0, t) = u(2\pi, t), u_\theta(0, t) = u_\theta(2\pi, t), & 0 \leq t \\ u(\theta, 0) = A \cos^2 \frac{\theta}{2}, & 0 \leq \theta < 2\pi \end{cases}$$

$$\lambda_n = n^2, \quad (n = 0, 1, 2, \dots)$$

$$\Phi_n(\theta) = \{1, \cos n\theta, \sin n\theta\}$$

习题



例

如图所示, 细金属丝被弯成半径为 ρ 的圆环, 经焊接制成环形线圈. 取焊接结束为0时刻, 此时圆环内的温度分布为 $A \cos^2 \frac{\theta}{2}$.

请写出描述该环形线圈温度演化的定解问题并求解

第3步 建立初值问题

$$u(\theta, t) = \sum_{n=0}^{\infty} T_n(t) \Phi_n(\theta) \quad \varphi(\theta) = \sum_{n=0}^{\infty} \varphi_n \cdot \Phi_n(\theta)$$

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}, & 0 \leq \theta < 2\pi, t > 0 \\ u(0, t) = u(2\pi, t), u_\theta(0, t) = u_\theta(2\pi, t), & 0 \leq t \\ u(\theta, 0) = A \cos^2 \frac{\theta}{2}, & 0 \leq \theta < 2\pi \end{cases}$$

代入原PDE

$$\frac{\partial u}{\partial t} - \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

$$\sum_{n=0}^{\infty} T'_n(t) \Phi_n(\theta) - \frac{a^2}{\rho^2} \sum_{n=0}^{\infty} T_n(t) \Phi''_n(\theta) = 0$$

$$\sum_{n=0}^{\infty} T'_n(t) \Phi_n(\theta) - \frac{a^2}{\rho^2} \sum_{n=0}^{\infty} T_n(t) \cdot [-\lambda_n \cdot \Phi_n(\theta)] = 0$$

$$\sum_{n=0}^{\infty} \left[T'_n(t) + \lambda_n \frac{a^2}{\rho^2} T_n(t) \right] \Phi_n(\theta) = 0$$

$$T'_n(t) + \lambda_n \frac{a^2}{\rho^2} T_n(t) = 0$$

$$\lambda_n = n^2, \quad (n = 0, 1, 2, \dots)$$

$$\Phi_n(\theta) = \{1, \cos n\theta, \sin n\theta\}$$

初始条件

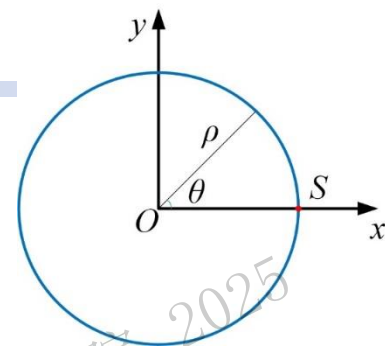
$$u(\theta, 0) = \sum_{n=0}^{\infty} T_n(0) \Phi_n(\theta) = \varphi(\theta) = \sum_{n=0}^{\infty} \varphi_n \cdot \Phi_n(\theta)$$

$$T_n(0) = \varphi_n$$

初值ODE

$$\begin{cases} T'_n(t) + \lambda_n \frac{a^2}{\rho^2} T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

习题



例

如图所示, 细金属丝被弯成半径为 ρ 的圆环, 经焊接制成环形线圈. 取焊接结束为0时刻, 此时圆环内的温度分布为 $A \cos^2 \frac{\theta}{2}$.

请写出描述该环形线圈温度演化的定解问题并求解

第4步 求解初值问题/回代

$$\begin{cases} T'_n(t) + \lambda_n \frac{a^2}{\rho^2} T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}, & 0 \leq \theta < 2\pi, t > 0 \\ u(0, t) = u(2\pi, t), u_\theta(0, t) = u_\theta(2\pi, t), & 0 \leq t \\ u(\theta, 0) = A \cos^2 \frac{\theta}{2}, & 0 \leq \theta < 2\pi \end{cases}$$

解得 $T_n(t) = C_1 \exp\left(-\lambda_n \frac{a^2}{\rho^2} t\right) = \varphi_n \exp\left(-\lambda_n \frac{a^2}{\rho^2} t\right)$

$$\lambda_n = n^2, \quad (n = 0, 1, 2, \dots)$$

$$\Phi_n(\theta) = \{1, \cos n\theta, \sin n\theta\}$$

原定解问题的解为 $u(\theta, t) = \sum_{n=0}^{\infty} T_n(t) \Phi_n(\theta) = \sum_{n=0}^{\infty} \varphi_n \exp\left(-\lambda_n \frac{a^2}{\rho^2} t\right) \Phi_n(\theta)$

$$= \varphi_0 \exp\left(-0 \cdot \frac{a^2}{\rho^2} t\right) \cdot 1 + \varphi_{1,\cos} \exp\left(-1^2 \cdot \frac{a^2}{\rho^2} t\right) \cos \theta$$

$$= \frac{A}{2} + \frac{A}{2} \exp\left(-\frac{a^2}{\rho^2} t\right) \cos \theta$$

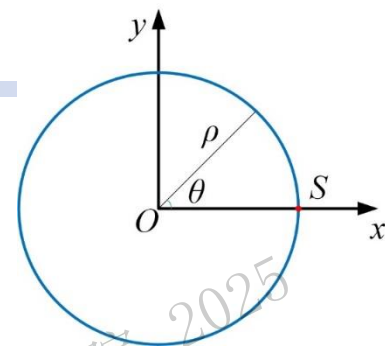
$$= \frac{A}{2} \left[1 + \exp\left(-\frac{a^2}{\rho^2} t\right) \cos \theta \right]$$

习题

例

如图所示, 细金属丝被弯成半径为 ρ 的圆环, 经焊接制成环形线圈. 取焊接结束为0时刻, 此时圆环内的温度分布为 $A \cos^2 \frac{\theta}{2}$.

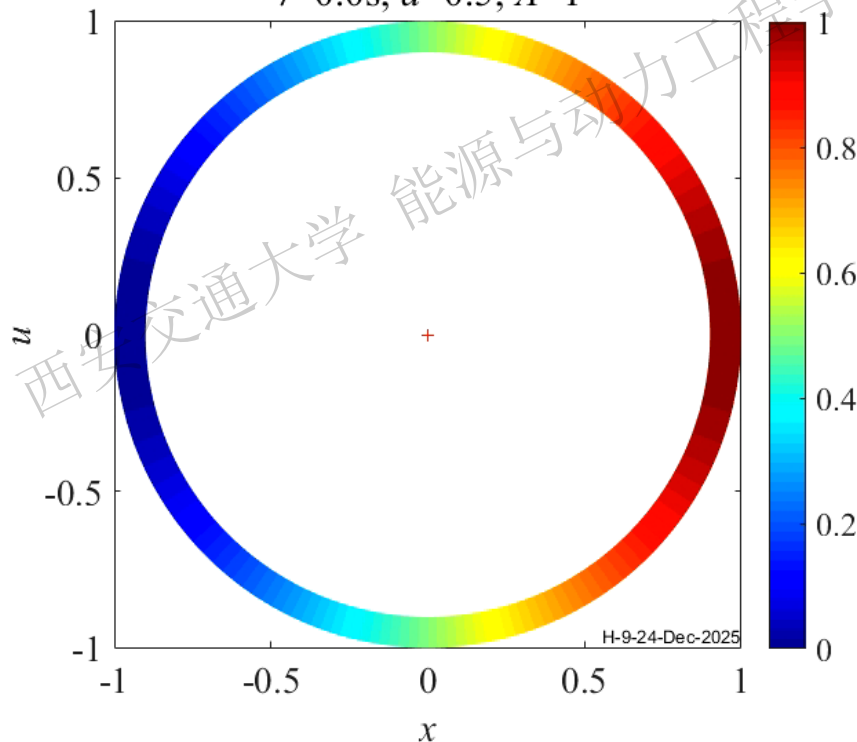
请写出描述该环形线圈温度演化的定解问题并求解



原定解问题的解为

$$u(\theta, t) = \frac{A}{2} \left[1 + \exp\left(-\frac{a^2}{\rho^2} t\right) \cos \theta \right]$$

$\tau=0.0s, a=0.5, A=1$



$$\begin{cases} \frac{\partial u}{\partial t} = \frac{a^2}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}, & 0 \leq \theta < 2\pi, t > 0 \\ u(0, t) = u(2\pi, t), u_\theta(0, t) = u_\theta(2\pi, t), & 0 \leq t \\ u(\theta, 0) = A \cos^2 \frac{\theta}{2}, & 0 \leq \theta < 2\pi \end{cases}$$

$$\lambda_n = n^2, \quad (n = 0, 1, 2, \dots)$$

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